

## Journal of Physics A: Mathematical and Theoretical

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## Exact Connectedness Threshold for Pseudospectra of the Open Hatano–Nelson Chain

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**Keywords:** open Hatano–Nelson chain, Euclidean pseudospectrum, non-Hermitian skin effect, connectedness threshold, least singular value, backward-error barrier

**Abstract**

We determine the exact finite-size transition at which the Euclidean pseudospectrum of the open Hatano–Nelson chain changes from separated mode-resolving components to a single connected uncertainty cloud. At fixed unstructured spectral-norm uncertainty, this merger marks the loss of component-based separation between open-boundary spectral labels. For the dimensionless one-particle matrix  $A_n(a) = \text{tridiag}(a, 0, 1)$ ,  $0 < a < 1$ , the least singular value of  $(x + iy)I_n - A_n(a)$  is nondecreasing in  $|y|$ ; consequently, every pseudospectral component is contractible. If  $\gamma_n$  is the largest real-axis least-singular-value barrier between adjacent open-boundary eigenvalues, then  $\sigma_\varepsilon(A_n)$  is connected if and only if  $\varepsilon > \gamma_n$ . We prove the strict finite-size monotonicity  $\gamma_{n+1} < \gamma_n$  and the sharp-order bounds  $\gamma_n \asymp_a (n+1)a^{n/2}$ . Thus the first-connected and eventual-connected thresholds coincide: once the pseudospectrum becomes connected, connectedness persists as the chain grows. For fixed  $a$ , the critical site count satisfies

$$N_c = \frac{2}{\log(1/a)} [\log(1/\varepsilon) + \log\log(1/\varepsilon)] + O_a(1) \quad (\varepsilon \downarrow 0).$$

The leading coefficient  $2/\log(1/a)$  is the amplitude skin depth in lattice-spacing units: the same imaginary gauge that localizes the open-boundary modes sets the asymptotic finite-size scale for the collapse of component-based worst-case spectral resolution.

**1 Introduction**

In a non-Hermitian open chain, fixed worst-case spectral uncertainty turns system size into a control parameter for mode resolution. The Hatano–Nelson model was introduced in [1, 2]. In the open chain considered here, a real spectrum coexists with strongly edge-localized eigenmodes. For a short clean chain, the Euclidean pseudospectrum consists of separated components, each containing one open-boundary eigenvalue. As the number of sites grows, nonnormal sensitivity expands these components until the last real-axis gap closes and a single connected uncertainty cloud remains. The finite-size problem is to determine the chain length of this final merger and establish whether connectedness persists as sites are added.

Connectedness defines a useful worst-case spectral-resolution transition because distinct pseudospectral components provide perturbation-independent cluster labels. Along any continuous perturbation path inside a prescribed spectral-norm ball, each eigenvalue branch remains confined to its initial component. After all components merge, spectral location alone ceases to supply such a label for the open-boundary modes. The pseudospectrum here is the union of spectra over the perturbation ball, whereas each fixed perturbation contributes only its own eigenvalues. Knowledge of the actual perturbation path may therefore permit branchwise tracking beyond the pseudospectral component merger.

With the site convention that  $t_R$  moves a particle from  $j$  to  $j+1$  and  $t_L$  moves it from  $j+1$  to  $j$ , take  $0 < t_R < t_L$ . A positive-sign one-particle representative of the

open chain is

$$H_n^{\text{OBC}} = t_L A_n(a), \quad a = \frac{t_R}{t_L},$$

where the dimensionless matrix is the real tridiagonal Toeplitz path

$$A_n(a) = \text{tridiag}(a, 0, 1) = \begin{bmatrix} 0 & 1 & & & \\ a & 0 & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & & a & 0 \end{bmatrix}, \quad n \geq 2, \quad 0 < a < 1, \quad (1.1)$$

where the three arguments of tridiag denote the subdiagonal, diagonal, and superdiagonal, respectively. For a matrix  $A \in \mathbb{C}^{n \times n}$ , the Euclidean  $\varepsilon$ -pseudospectrum under the strict convention used here is

$$\sigma_\varepsilon(A) = \sigma(A) \cup \{z \in \mathbb{C} : \|(zI_n - A)^{-1}\|_2 > \varepsilon^{-1}\} = \{z \in \mathbb{C} : s_{\min}(zI_n - A) < \varepsilon\}. \quad (1.2)$$

Equivalently, it is the union of the spectra of  $A + \Delta A$  over all unstructured complex perturbations with  $\|\Delta A\|_2 < \varepsilon$  [3]. The strict inequalities in this convention are important at the merger threshold. For nonnormal matrices, pseudospectra also expose spectral sensitivity and resolvent growth that the eigenvalues alone can miss [3, 4].

The main result is most simply described by a real-axis backward-error barrier. Let  $\gamma_n$  be the largest, over all gaps between adjacent open-boundary eigenvalues, of the maximum least singular value  $s_{\min}(xI_n - A_n)$  in that gap. Thus  $\gamma_n$  is the largest backward error that must be crossed to remove the final separation between adjacent open-boundary spectral labels. We prove that every pseudospectral component is contractible and that  $\sigma_\varepsilon(A_n)$  is connected exactly when  $\varepsilon > \gamma_n$ . The decisive finite-size result is  $\gamma_{n+1} < \gamma_n$  for every  $n \geq 2$ : a fixed uncertainty has one system-size crossing, after which the cloud remains connected. The sharp scale  $\gamma_n \asymp_a (n+1)a^{n/2}$  yields

$$N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log \log(1/\varepsilon)) + O_a(1).$$

The coefficient  $2/\log(1/a)$  is the amplitude skin depth; the intensity decay length is half as large. Thus the result is global in spectral parameter and exact at finite size: it determines the topology of the complete Euclidean pseudospectrum, identifies the final inter-mode barrier, proves its strict decrease with system size, and derives the skin-depth-controlled site threshold.

The theorem concerns unstructured spectral-norm uncertainty and identifies the merger of the complete finite-dimensional Euclidean pseudospectrum into one spectral-resolution class. The non-Hermitian skin effect is already present in the eigenmodes at every  $n$ . Single-realization spectra, structured-disorder thresholds, and dynamical amplification are complementary problems that may be governed by different scales.

The imaginary gauge makes the physical origin of the exponential scale transparent. Put

$$h = -\frac{1}{2} \log a > 0, \quad a = e^{-2h}, \quad r = \sqrt{a} = e^{-h}.$$

The diagonal gauge

$$\mathcal{G}_n(\varrho) = \text{diag}(1, \varrho, \dots, \varrho^{n-1})$$

and the upper-shift matrix  $V_n = \text{tridiag}(0, 0, 1)$  give the dimensionless similarity

$$A_n(a) = \mathcal{G}_n(r) [r(V_n + V_n^T)] \mathcal{G}_n(r)^{-1}.$$

Multiplying by  $t_L$  and using  $t_L r = \sqrt{t_L t_R}$  recovers the usual physical imaginary-gauge formula for  $H_n^{\text{OBC}}$ . The nonunitary similarity changes the singular-value geometry underlying the Euclidean pseudospectrum. The open-boundary eigenvalues are real, while the right eigenmodes carry the envelope  $e^{-h(j-1)}$  and accumulate at the left edge. This is the non-Hermitian skin effect [5, 6, 7]; see [8] for a review. Under periodic boundary conditions the Bloch dispersion traces

$$\mathcal{E}_{\text{PBC}}(k) = t_L e^{ik} + t_R e^{-ik} = (t_L + t_R) \cos k + i(t_L - t_R) \sin k.$$

A finite periodic ring samples this ellipse at  $k = 2\pi q/n$ ,  $q = 0, \dots, n-1$ . The connectedness transition studied here asks how the same nonreciprocal gauge controls the worst-case resolution of the finite open-boundary spectrum.

The analysis assumes  $0 < a < 1$ . At  $a = 0$ ,  $A_n(0) = V_n$  is the unidirectional Jordan-block limit, the diagonal gauge is singular, and the limits  $a \downarrow 0$  and  $\delta \downarrow 0$ , where  $\delta$  is the normalized uncertainty level used below, can depend on their order. Accordingly, the fixed- $a$  small-uncertainty asymptotic is stated for  $0 < a < 1$ .

The relevant mathematical literature has several distinct strands. Classical limit-set theory for banded Toeplitz matrices was initiated in [9]; later work developed inverse-norm, spectral, and finite-section pseudospectral theory in [10, 11, 12, 13]. Closed-form spectral, conditioning, and pseudospectral formulas for tridiagonal Toeplitz matrices are surveyed in [14]. For general matrices, component coalescence and distance to defectivity are linked in [15]; optimization and variational analyses of pseudospectra appear in [16, 17]. These works provide broad spectral and geometric theory. The consecutive-size comparison of the global barrier established here addresses a distinct finite-size problem.

Pseudospectral approaches to Hatano–Nelson-type localization include random bidiagonal models [18] and interpretations of nonnormal boundary sensitivity [19, 20]. Mathematical foundations of the skin effect were developed in [21]. Spectra and pseudospectra of tridiagonal  $k$ -Toeplitz matrices were analysed in [22], with the finite-section pseudospectral convergence statement and the resulting limiting-spectrum theorems corrected in the published corrigendum [23]. These contributions concern broader Toeplitz or boundary-sensitivity mechanisms. The present result addresses a separate finite-dimensional connectedness problem.

For asymmetric lattices, Kiorpelidis and Makris [24] reported numerically an eigenvalue-cloud merger crossover, denoted  $\widetilde{N}$ , and distinguished it from the persistence of long-time dynamical amplification. Sirker [25] analyses open-chain pseudospectral contours and, at a fixed reference energy  $E$ , the exponentially small singular-value branch tied to the Toeplitz index. Here the object is the topology of the complete finite-dimensional Euclidean pseudospectrum at fixed uncertainty. We identify its global inter-eigenvalue barrier and prove that the barrier decreases strictly with matrix size, thereby excluding re-entrant splitting. The same barrier yields the finite-size threshold and its logarithmic-plus-log-log asymptotic.

The proof has three conceptual stages. Vertical monotonicity first reduces the planar connectedness problem to real-axis barriers. Persymmetry then converts singular values into signed eigenvalue branches; folded Chebyshev coordinates compare the noncentral gaps, while interlacing resolves the parity-changing central gaps. Finally, the topology argument and sharp barrier bounds yield the exact connectedness criterion and the skin-depth-controlled threshold law. The algebraic certificates required by these steps are recorded in [appendices A to F](#). The figures in [section 6](#) illustrate the transition, barrier scale, and skin-depth mechanism, and [section 7](#) restores the dimensional Hatano–Nelson interpretation.

## 2 Model and main results

This section turns the merger problem into an exact comparison between the uncertainty level and the largest real-axis barrier. The resulting scalar quantity retains the global physical content of the pseudospectrum: it measures the perturbation needed to remove the final separation between adjacent open-boundary spectral labels.

For  $n \geq 2$  and  $0 < a < 1$ , let

$$A_n(a) = \text{tridiag}(a, 0, 1).$$

When  $a$  is fixed, we write  $A_n$  for  $A_n(a)$ . For  $\varepsilon > 0$ , write

$$\sigma_\varepsilon(A_n) = \{z \in \mathbb{C} : s_{\min}(zI_n - A_n) < \varepsilon\}.$$

Define

$$N_e(a, \varepsilon) = \min\{N \geq 2 : \sigma_\varepsilon(A_n) \text{ is connected for every } n \geq N\},$$

$$N_f(a, \varepsilon) = \min\{N \geq 2 : \sigma_\varepsilon(A_N) \text{ is connected}\}.$$

The following theorem proves that both defining sets are nonempty.

For the remainder of the article, put

$$r = \sqrt{a}, \quad J_n = [\mathbf{1}_{\{i+j=n+1\}}]_{i,j=1}^n.$$

Thus  $A_n = V_n + aV_n^T$ . Here,  $\mathbf{e}_j$  denotes a Euclidean coordinate vector of the ambient dimension, and  $U_j(t)$  denotes the Chebyshev polynomial of the second kind, with  $U_{-1} = 0$ .

When the dimension is  $n$ , write  $N = n + 1$  and define

$$I_n^{\text{sp}} = \left[-2r \cos \frac{\pi}{N}, 2r \cos \frac{\pi}{N}\right], \quad g_n(x) = s_{\min}(xI_n - A_n), \quad \gamma_n = \max_{x \in I_n^{\text{sp}}} g_n(x). \quad (2.1)$$

The explicit lower and upper barriers used below are

$$\mathcal{U}_n(a) = r^n \csc \frac{\pi}{N}, \quad (2.2)$$

$$\mathcal{L}_n(a) = \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right) \frac{r^n(1 - r^2)}{(1 - r^{2N}) \sin \frac{3\pi}{2N}}. \quad (2.3)$$

The quantity  $g_n(x)$  is the minimum unstructured spectral-norm perturbation needed to place an eigenvalue at the real point  $x$ . Accordingly,  $\gamma_n$  will be shown to be the largest backward-error barrier separating adjacent open-boundary spectral labels. It is a global quantity: determining it requires examining every real inter-eigenvalue gap, whereas one spectral parameter provides only local information.

**Theorem 2.1** (Main theorem). *For every  $0 < a < 1$  and  $\varepsilon > 0$ , the following assertions hold.*

1. *For every  $n \geq 2$ , every connected component of  $\sigma_\varepsilon(A_n(a))$  is contractible.*
2. *For  $n \geq 2$ , the barrier sequence is strictly decreasing,  $\gamma_{n+1} < \gamma_n$ , tends to zero, and*

$$\sigma_\varepsilon(A_n(a)) \text{ is connected} \iff \varepsilon > \gamma_n.$$

3. The first-connectedness and eventual thresholds coincide:

$$N_c(a, \varepsilon) := N_f(a, \varepsilon) = N_e(a, \varepsilon).$$

Moreover,

$$1 + \max\{n \geq 2 : \mathcal{L}_n(a) \geq \varepsilon\} \leq N_c(a, \varepsilon) \leq \min\{n \geq 2 : \mathcal{U}_n(a) < \varepsilon\}, \quad (2.4)$$

where the maximum of the empty set is understood to be 1.

4. For fixed  $a$ , as  $\varepsilon \downarrow 0$ ,

$$N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log\log(1/\varepsilon)) + O_a(1), \quad (2.5)$$

and  $N_c(a, \varepsilon) = 2$  if and only if  $\varepsilon > a$ .

The theorem turns the growth-induced cloud merger into an exact size-dependent transition. Vertical monotonicity makes every component connection visible on the real energy axis, where  $\gamma_n$  is the final backward-error barrier between open-boundary spectral labels. Strict comparison in consecutive dimensions makes the transition one-way, and the sharp-order bounds show that the imaginary gauge controls both skin localization and the loss of global spectral-cluster resolution.

### 3 Vertical monotonicity

The topological reduction rests on a monotonicity property that makes every pseudospectral component connection visible on the real axis.

**Lemma 3.1** (Vertical monotonicity). *For every real  $x$ ,*

$$y \mapsto s_{\min}((x + iy)I_n - A_n), \quad y \geq 0,$$

*is nondecreasing.*

*Proof.* Put  $Y = y^2$ , and, for  $t \geq 0$ , define

$$\mathcal{P}_n(Y, t) = ((x + i\sqrt{Y})I_n - A_n)^*((x + i\sqrt{Y})I_n - A_n) - t^2 I_n, \quad F_n(Y, t) = \det \mathcal{P}_n(Y, t).$$

For each  $k \geq 1$ , define  $\mathcal{P}_k(Y, t)$  by replacing  $n$  with  $k$ , and set  $F_k(Y, t) = \det \mathcal{P}_k(Y, t)$ ; also set  $F_0 = 1$ . The key finite-dimensional implication is

$$\mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0. \quad (3.1)$$

The determinant implication (3.1) is the only algebraic input needed for the continuation argument. Its proof is given in [proposition A.1](#). The endpoint-corrected pentadiagonal pencil leads to a convolution of positive principal minors. Only one convolution coefficient can be negative, and a strict cofactor inequality absorbs it with the positive surplus  $2a^2(1 - a)^2$ .

Finally fix  $Y_0 \geq 0$ . When the least singular value at  $Y_0$  equals zero, monotonicity follows from nonnegativity of singular values. Otherwise take  $0 \leq t < s_{\min}((x + i\sqrt{Y_0})I_n - A_n)$ . On the maximal interval to the right of  $Y_0$  on which  $\mathcal{P}_n(Y, t) > 0$ , implication (3.1) makes its determinant strictly increasing. At any finite right endpoint, continuity of Hermitian eigenvalues would produce a positive-semidefinite singular matrix with determinant zero. Strict increase from the positive determinant at  $Y_0$  precludes such an endpoint. Thus  $\mathcal{P}_n(Y, t) > 0$  for every  $Y \geq Y_0$ . Letting  $t$  increase to the least singular value at  $Y_0$  proves the lemma.  $\square$

Consequently, at fixed real part, every vertical section of the strict pseudospectrum is an interval symmetric about the real axis. This is the topological reduction used in [section 5](#): every component connection is visible on the real trace, so real-axis barriers determine component separation.

#### 4 Strict finite-size comparison of real-gap barriers

The final barrier can occur in any inter-eigenvalue gap, and the identity of the central gap changes with parity. A global finite-size comparison must therefore control every gap. Persymmetry turns singular values into signed eigenvalue branches, folded Chebyshev coordinates parametrize those branches, scale inequalities compare noncentral gaps, and a separate interlacing argument handles the parity-changing central gaps. We first define the local real-gap barriers. Write the eigenvalues in decreasing order as

$$\lambda_{n,k} = 2r \cos \frac{k\pi}{n+1}, \quad 1 \leq k \leq n,$$

and call

$$G_{n,k} = [\lambda_{n,k+1}, \lambda_{n,k}], \quad \gamma_{n,k} = \max_{x \in G_{n,k}} g_n(x), \quad 1 \leq k < n,$$

the closed spectral gaps and their heights. Continuity guarantees that the maximum on each compact gap is attained, while the spectral endpoints have height zero. Every interior gap point belongs to the resolvent set, so  $xI_n - A_n$  is invertible there and  $g_n(x) > 0$ . Consequently every gap height is positive and every maximizer lies in the open gap. Hence

$$\gamma_n = \max_{1 \leq k < n} \gamma_{n,k}. \quad (4.1)$$

Thus  $\gamma_{n,k}$  is the local backward-error barrier between two adjacent open-boundary modes, and  $\gamma_n$  is the final such barrier to be overcome. Strict decrease of this maximum makes the connectedness transition persistent as the site count grows.

**Lemma 4.1** (Strict barrier comparison). *For every  $n \geq 2$ ,*

$$\gamma_{n+1} < \gamma_n.$$

Persymmetry supplies the signed spectral representation needed for the finite-size comparison:  $A_n J_n = J_n A_n^T$ , so  $B_n(x) = (xI_n - A_n)J_n$  is real symmetric. Moreover,

$$(xI_n - A_n - sJ_n)J_n = B_n(x) - sI_n.$$

Since  $J_n$  is orthogonal, the absolute eigenvalues of  $B_n(x)$  are exactly the singular values of  $xI_n - A_n$ , including multiplicity. Consequently,

$$s_{\min}(xI_n - A_n) = \min\{|s| : \det(xI_n - A_n - sJ_n) = 0\}. \quad (4.2)$$

Let  $q_n(x, s) = \det(xI_n - A_n - sJ_n)$ .

**Lemma 4.2** (The middle signed branch). *Write the eigenvalues of the real symmetric matrix*

$$B_n(x) = (xI_n - A_n)J_n$$

*in decreasing order as  $\alpha_1(x) \geq \dots \geq \alpha_n(x)$ , and abbreviate  $\lambda_k = \lambda_{n,k}$ . If  $\lambda_{j+1} < x < \lambda_j$ , then*

$$s_{\min}(xI_n - A_n) = |s_*(x)|, \quad \operatorname{sgn}(s_*(x)) = (-1)^j,$$

*where*

$$s_*(x) = \begin{cases} \alpha_m(x), & n = 2m, \\ \alpha_{m+1}(x), & n = 2m + 1. \end{cases}$$

*Proof.* Let  $A_n^{\text{sym}} = r(V_n + V_n^T)$ . Since  $A_n = \mathcal{G}_n(r)A_n^{\text{sym}}\mathcal{G}_n(r)^{-1}$  and  $\mathcal{G}_n(r)^{-1}J_n = r^{-(n-1)}J_n\mathcal{G}_n(r)$ ,

$$B_n(x) = r^{-(n-1)}\mathcal{G}_n(r)(xI_n - A_n^{\text{sym}})J_n\mathcal{G}_n(r). \quad (4.3)$$

The sine eigenvector of  $A_n^{\text{sym}}$  belonging to  $\lambda_k$  is an eigenvector of  $J_n$  with eigenvalue  $(-1)^{k+1}$ . Hence (4.3) and Sylvester's law of inertia show that  $B_n(x)$  has the same inertia as the diagonal matrix with entries

$$(-1)^{k+1}(x - \lambda_k), \quad 1 \leq k \leq n.$$

Counting signs gives

$$\nu_+(B_{2m}(x)) = \begin{cases} m, & j \text{ even,} \\ m-1, & j \text{ odd,} \end{cases} \quad \nu_+(B_{2m+1}(x)) = \begin{cases} m+1, & j \text{ even,} \\ m, & j \text{ odd.} \end{cases} \quad (4.4)$$

First let  $n = 2m$ . Permute the coordinates into the two groups

$$(1, \dots, m), \quad (2m, 2m-1, \dots, m+1).$$

In these coordinates

$$\begin{aligned} B_{2m}(x) &= \hat{P}_m(x) - \mathcal{R}_{\text{bd}}, \\ \hat{P}_m(x) &= \begin{bmatrix} 0 & P_m(x) \\ P_m(x)^T & 0 \end{bmatrix}, \\ \mathcal{R}_{\text{bd}} &= \begin{bmatrix} \mathbf{e}_m \mathbf{e}_m^T & 0 \\ 0 & a \mathbf{e}_m \mathbf{e}_m^T \end{bmatrix}, \end{aligned} \quad (4.5)$$

where  $P_m(x) = xI_m - A_m$ . Thus  $\mathcal{R}_{\text{bd}} \geq 0$  and  $\text{rank } \mathcal{R}_{\text{bd}} = 2$ . For  $\Gamma = \text{diag}(I_m, -I_m)$ ,

$$\tilde{B} = -\Gamma B_{2m}(x) \Gamma = \hat{P}_m(x) + \mathcal{R}_{\text{bd}} = B_{2m}(x) + 2\mathcal{R}_{\text{bd}}.$$

If  $\tilde{\alpha}_1 \geq \dots \geq \tilde{\alpha}_{2m}$  are the eigenvalues of  $\tilde{B}$ , then

$$\tilde{\alpha}_k = -\alpha_{2m+1-k}. \quad (4.6)$$

Positive-semidefinite rank-two interlacing gives

$$\tilde{\alpha}_k \geq \alpha_k \geq \tilde{\alpha}_{k+2}, \quad 1 \leq k \leq 2m-2. \quad (4.7)$$

If  $j$  is even, the only candidates for minimum modulus have moduli  $\alpha_m$  and  $-\alpha_{m+1} = \tilde{\alpha}_m$ , and  $\alpha_m \leq \tilde{\alpha}_m$ . If  $j$  is odd and  $m \geq 2$ , the two candidates have moduli  $\alpha_{m-1}$  and  $-\alpha_m = \tilde{\alpha}_{m+1}$ ; taking  $k = m-1$  in (4.7) gives  $\alpha_{m-1} \geq \tilde{\alpha}_{m+1}$ . For  $m = 1$ , both eigenvalues in the sole gap are negative, so the larger one  $\alpha_1$  has minimum modulus. Thus  $\alpha_m$  always attains the minimum modulus, and its sign follows from (4.4). Equality in an interlacing comparison is harmless: it only gives a multiple least singular value.

Now let  $n = 2m+1$ . Group the first and last  $m$  coordinates as above and leave the middle coordinate last. Then

$$B_{2m+1}(x) = \begin{bmatrix} \hat{P}_m(x) & \mathbf{w} \\ \mathbf{w}^T & x \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -\mathbf{e}_m \\ -a\mathbf{e}_m \end{bmatrix}. \quad (4.8)$$

In an open spectral gap of  $A_{2m+1}$ ,  $P_m(x)$  is nonsingular. Indeed, the explicit path spectrum gives

$$\text{spec}(A_m) = \left\{ 2r \cos \frac{\ell\pi}{m+1} : 1 \leq \ell \leq m \right\} = \{ \lambda_{2m+1, 2\ell} : 1 \leq \ell \leq m \},$$

so its zeros are exactly the even-indexed eigenvalues of  $A_{2m+1}$ , none of which lies in an open spectral gap. The eigenvalues of the principal submatrix  $\hat{P}_m(x)$  are consequently

$$s_1(P_m) \geq \cdots \geq s_m(P_m) > 0 > -s_m(P_m) \geq \cdots \geq -s_1(P_m).$$

Cauchy interlacing gives

$$\alpha_m \geq s_m(P_m) \geq \alpha_{m+1} \geq -s_m(P_m) \geq \alpha_{m+2}.$$

If  $j$  is even, (4.4) and interlacing give

$$0 < \alpha_{m+1} \leq s_m(P_m) \leq -\alpha_{m+2};$$

if  $j$  is odd, they give

$$0 < -\alpha_{m+1} \leq s_m(P_m) \leq \alpha_m.$$

Every other eigenvalue is farther from zero. Hence  $\alpha_{m+1}$  attains the minimum modulus, with sign prescribed by (4.4); equality again only means multiplicity. At a spectral endpoint the zero is simple because  $A_n$  is similar to the Jacobi matrix  $A_n^{\text{sym}}$ , whose eigenvalues are simple. Thus the selected ordered branch extends continuously to zero at both ends of the closed gap. Finally, multiplication by the orthogonal matrix  $J_n$  preserves singular values, which proves the assertion.  $\square$

*Proof of Lemma 4.1.* With the signed branch in hand, we introduce folded coordinates. They compare the noncentral gaps, resolve the central pair, and assemble the global inequality.

To avoid ambiguity,  $F_n$  continues to denote the determinant from the vertical argument, whereas  $\mathcal{F}_h$  below is an angular factor; the fraktur quantities  $\mathfrak{c}_h, \mathfrak{d}_h$  are distinct from the central gap heights  $c_m, d_m$ .

*Folded continuants*

Put

$$C_0 = x^2 + (1-a)^2 - s^2, \quad \zeta_s = \frac{1+a^2 + (1+a)s}{2a},$$

and define the folded discriminant and recurrence coefficient by

$$\Delta_{\text{fold}} = ((1+a-x)^2 - s^2)((1+a+x)^2 - s^2),$$

$$\mathfrak{b} = (1+a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3.$$

The global recurrence polynomial is

$$\begin{aligned} \Pi(z) &:= 1 - C_0 z + \mathfrak{b} z^2 - a^2 C_0 z^3 + a^4 z^4, \\ \left. \begin{aligned} 2a(u+v) &= C_0, \\ a^2(4uv+2) &= \mathfrak{b} \end{aligned} \right\} &\implies \Pi(z) = (1 - 2auz + a^2 z^2)(1 - 2avz + a^2 z^2), \quad u, v \in \mathbb{R}. \end{aligned} \tag{4.9}$$

On a real sheet with  $\Delta_{\text{fold}} \geq 0$ , we additionally put

$$\xi_{\pm} = \frac{C_0 \pm \sqrt{\Delta_{\text{fold}}}}{4a}.$$

Then the pair  $(u, v) = (\xi_+, \xi_-)$  satisfies the two symmetric relations in (4.9):

$$\xi_+ + \xi_- = \frac{C_0}{2a}, \quad \mathfrak{b} = a^2(4\xi_+\xi_- + 2).$$

We use the root presentation on these real sheets and the polynomial  $\Pi$  itself for the global transfer and generating identities. The complete five-minor transfer, characteristic-polynomial calculation, boundary residuals, and generating functions are given in [appendix B](#). Their parity-unified conclusion is the part needed in the main argument. For a differentiable real function  $F$ , write

$$[F]_{u,v} := \begin{cases} \frac{F(u) - F(v)}{u - v}, & u \neq v, \\ F'(u), & u = v \end{cases}.$$

Whenever the real pair  $u, v$  satisfies the two symmetric relations in (4.9), the divided-difference calculation in [appendix C](#) gives

$$q_{2m}(x, s) = a^m [\mathbf{f}_m]_{u,v}, \quad \mathbf{f}_m(t) = (t - \zeta_s)(U_m(t) + U_{m-1}(t)), \quad (4.10)$$

and

$$q_{2m+1}(x, s) = a^m [\mathbf{g}_m]_{u,v}, \quad \mathbf{g}_m(t) = \left( (x - s)t - \frac{x(1 + a^2) + 2as}{2a} \right) U_m(t). \quad (4.11)$$

The recurrence and generating identities are polynomial and therefore global. The real-pair formulas are used only after the two symmetric relations have been checked; at  $u = v$ , the displayed derivative is the collision convention. For  $\Delta_{\text{fold}} < 0$ , the argument stays at the polynomial level and avoids any complex square-root branch.

These formulas replace the singular-value problem by a signed scalar branch with parity already resolved. The next step identifies a geometric parametrization of that branch.

#### Half-angle chord parametrization

Recall that  $a = e^{-2h}$ , and put

$$\mathbf{s}_h = \sinh h, \quad \mathbf{c}_h = \cosh h.$$

For  $K \geq 2$  and  $0 \leq \theta \leq \pi$ , define

$$\Phi_K(\theta) = \sqrt{\mathbf{s}_h^2 + \sin^2 \theta} \frac{|\sin(K\theta)|}{\sin \theta}, \quad (4.12)$$

where the quotient is interpreted continuously at  $\theta = 0, \pi$ , and

$$\Psi_K(\eta) = \sqrt{\mathbf{s}_h^2 - \sinh^2 \eta} \frac{\sinh(K\eta)}{\sinh \eta}, \quad 0 \leq \eta \leq h,$$

with the limiting value at  $\eta = 0$ . The following half-angle reduction also fixes the sign convention. Put

$$y = \cos \theta, \quad z = \cosh \eta, \quad \mathbf{p}_\theta = \sqrt{\mathbf{c}_h^2 - y^2}, \quad \mathbf{p}_\eta = \sqrt{\mathbf{c}_h^2 - z^2},$$

and let  $L_0 = 2r/\mathbf{c}_h$ . If  $x = L_0 y z$ ,  $s = L_0 \mathbf{p}_\theta \mathbf{p}_\eta > 0$ , and  $\tau = \varsigma s$  with  $\varsigma \in \{-1, 1\}$ , then direct expansion gives

$$\begin{aligned} (1 + a + x)^2 - s^2 &= 4a(z + y)^2, & (1 + a - x)^2 - s^2 &= 4a(z - y)^2, \\ u = 2z^2 - 1, & v = 2y^2 - 1, & \zeta_\tau &= 2\mathbf{c}_h^2 - 1 + 2\varsigma \mathbf{p}_\theta \mathbf{p}_\eta. \end{aligned} \quad (4.13)$$

The real pair  $(u, v)$  satisfies the two symmetric relations in (4.9). For  $K = 2m + 1$ , use

$$U_m(2t^2 - 1) + U_{m-1}(2t^2 - 1) = U_{K-1}(t);$$

for  $K = 2m + 2$ , use

$$U_m(2t^2 - 1) = \frac{U_{K-1}(t)}{2t}.$$

Substitution of this pair in (4.10) and (4.11), respectively, reduces either equation  $q_{K-1}(x, \tau) = 0$  to

$$\mathfrak{p}_\theta U_{K-1}(y) = \varsigma \mathfrak{p}_\eta U_{K-1}(z). \quad (4.14)$$

In the even-fold case the intermediate factorization is

$$(\mathfrak{p}_\eta + \varsigma \mathfrak{p}_\theta) (\mathfrak{p}_\eta U_{K-1}(z) - \varsigma \mathfrak{p}_\theta U_{K-1}(y)) = 0.$$

The first factor could vanish only if  $\varsigma = -1$  and  $\mathfrak{p}_\eta = \mathfrak{p}_\theta$ , which would imply  $y^2 = z^2$ ; the distinct-variable condition therefore leaves precisely (4.14). In the odd-fold case the reduction also contains the factor  $\mathfrak{p}_\eta y + \varsigma \mathfrak{p}_\theta z$ . On a distinct-variable sheet this factor is nonzero: if it vanished, squaring would give

$$(\mathfrak{c}_h^2 - z^2)y^2 = (\mathfrak{c}_h^2 - y^2)z^2,$$

and hence  $y^2 = z^2$ . Thus the factor is nonzero whenever the folded variables are distinct. On the outer lobe  $U_{K-1}(z) > 0$ , whereas, for  $j\pi/K < \theta < (j+1)\pi/K$ ,

$$\text{sgn}(U_{K-1}(\cos \theta)) = \text{sgn}(\sin(K\theta)) = (-1)^j.$$

Moreover,  $\mathfrak{p}_\theta |U_{K-1}(y)| = \Phi_K(\theta)$  and  $\mathfrak{p}_\eta U_{K-1}(z) = \Psi_K(\eta)$ . Thus (4.14) is equivalent to  $\varsigma = (-1)^j$  and

$$\Phi_K(\theta) = \Psi_K(\eta), \quad (4.15)$$

with

$$s = \mathfrak{s}_{\text{hyp}}(\theta, \eta) := \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 - \sinh^2 \eta)}, \quad x = \frac{2r}{\mathfrak{c}_h} \cos \theta \cosh \eta. \quad (4.16)$$

Consequently  $q_{K-1}(x, (-1)^j s) = 0$ . Conversely, for distinct real folded variables, their sum and product determine

$$x^2 = L_0^2 y^2 z^2, \quad s^2 = L_0^2 (\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2).$$

Thus, for distinct real variables and a fixed sign of  $x$ , the preceding steps are reversible, and the two chord relations are equivalent to the corresponding signed equation. The determinant identities themselves extend to a folded-variable collision by the continuous divided-difference convention, but we invoke the chord equivalence only for distinct variables; the selected middle branch is proved below to be collision-free. Here  $K - 1$  is the matrix size. If the exterior argument has crossed through 1, write it as  $z = \cos \phi$ ,  $0 \leq \phi < \pi/K$ . Then the chord equation is  $\Phi_K(\theta) = \Phi_K(\phi)$ , and the second factor in the formula for  $s$  is  $\mathfrak{s}_h^2 + \sin^2 \phi$ . We denote the resulting value by

$$\mathfrak{s}_{\text{ell}}(\theta, \phi) = \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 + \sin^2 \phi)}.$$

#### Branch classification and outer-lobe geometry

It remains to identify which chords realize the middle branch of Lemma 4.2. Set

$$\mathcal{W}_K(u) = \sqrt{\mathfrak{c}_h^2 - u^2} |U_{K-1}(u)|, \quad |u| \leq \mathfrak{c}_h. \quad (4.17)$$

At  $u = \pm \mathbf{c}_h$ , this denotes the continuous endpoint value 0. Then

$$\Phi_K(\theta) = \mathcal{W}_K(\cos \theta), \quad \Psi_K(\eta) = \mathcal{W}_K(\cosh \eta).$$

If  $u_{\ell,K}^{(0)} = \cos \frac{\ell\pi}{K}$ , then, away from the zeros,

$$\frac{d}{du} \log \mathcal{W}_K(u) = \sum_{\ell=1}^{K-1} \frac{1}{u - u_{\ell,K}^{(0)}} - \frac{u}{\mathbf{c}_h^2 - u^2},$$

and its derivative is

$$-\sum_{\ell=1}^{K-1} \frac{1}{(u - u_{\ell,K}^{(0)})^2} - \frac{\mathbf{c}_h^2 + u^2}{(\mathbf{c}_h^2 - u^2)^2} < 0. \quad (4.18)$$

At each finite nodal endpoint the logarithmic derivative tends to the appropriate signed infinity, and it tends to  $-\infty$  as  $u \uparrow \mathbf{c}_h$ . Thus there is exactly one lobe maximum in every nodal interval, including the outer interval  $\mathcal{J}_K^{\text{out}} = (\cos \frac{\pi}{K}, \mathbf{c}_h)$ . Let  $u_K^{\text{max}}$  be the unique maximizer on  $\mathcal{J}_K^{\text{out}}$ . For every  $0 \leq \ell < \mathcal{W}_K(u_K^{\text{max}})$ , there is a unique solution

$$z_K(\ell) \in (u_K^{\text{max}}, \mathbf{c}_h], \quad \mathcal{W}_K(z_K(\ell)) = \ell;$$

we call it the *endpoint-side outer solution*. If  $z_K(\ell) \geq 1$ , write it as  $\cosh \eta$ ,  $0 \leq \eta \leq h$ ; if it is at most 1, write it as  $\cos \phi$ ,  $0 \leq \phi < \pi/K$ . These are two coordinates on the same outer lobe, joined at  $z = 1$ .

For a positive-half gap and  $\theta \in (j\pi/K, (j+1)\pi/K)$ ,  $\theta \leq \pi/2$ , a *j-chord* is the pair

$$y = \cos \theta, \quad z = z_K(\mathcal{W}_K(y)), \quad \mathcal{W}_K(y) = \mathcal{W}_K(z),$$

together with the coordinate-free values

$$x = L_0 y z, \quad s = L_0 \sqrt{(\mathbf{c}_h^2 - y^2)(\mathbf{c}_h^2 - z^2)};$$

its signed root is  $(-1)^j s$ . The term endpoint-side always refers to the part of the outer lobe adjacent to  $\mathbf{c}_h$ , independently of whether it is described by the hyperbolic or elliptic coordinate.

Reflection reduces the branch classification to a positive-half gap, so  $j \geq 1$  and  $\theta \leq \pi/2$ . Put  $\phi_{\text{rem}} = \theta - j\pi/K \in (0, \pi/K)$ . Since  $|\sin(K\phi_{\text{rem}})| = |\sin(K\theta)|$  and

$$\mathcal{F}_h(t) = \frac{\sqrt{\mathbf{s}_h^2 + \sin^2 t}}{\sin t}$$

is strictly decreasing on  $(0, \pi/2]$ ,

$$\Phi_K(\theta) < \Phi_K(\phi_{\text{rem}}) \leq \max_{u \in \mathcal{J}_K^{\text{out}}} \mathcal{W}_K(u). \quad (4.19)$$

Thus the endpoint-side outer solution exists for every inner parameter.

At  $x = \lambda_{K-1,k} = 2r \cos \frac{k\pi}{K}$ ,  $s = 0$ , the two folded variables are  $\cos \frac{2k\pi}{K}$  and  $\cosh(2h)$ , or, in half variables,  $\cos \frac{k\pi}{K}$  and  $\mathbf{c}_h$ . Moreover, zero is a simple eigenvalue of  $B_{K-1}(\lambda_{K-1,k})$ : the matrix  $\lambda_{K-1,k} I_{K-1} - A_{K-1}$  has rank  $K-2$ , and  $J_{K-1}$  is invertible. The implicit-function theorem therefore gives, near either gap endpoint, a real continuation of the middle signed root and real half variables  $y = \cos \theta$  and  $z$ .

### Chord-sheet exhaustion

The endpoint germ extends through the whole positive open gap on the endpoint-side chord sheet. More precisely, if

$$\frac{j\pi}{K} < \theta < \frac{(j+1)\pi}{K}, \quad \frac{(j+1)\pi}{K} \leq \frac{\pi}{2},$$

then every point of the ordered middle branch is represented by the selected endpoint-side  $j$ -chord, and every such chord parameter produces a point of that branch. Its signed root is  $(-1)^j s$ , its magnitude is  $s_{\min}(xI_{K-1} - A_{K-1})$ , and the folded variables remain distinct in the open gap. When  $K$  is odd, the same statement holds on the positive half of the single central gap up to  $\theta = \pi/2$ , with reflection covering the negative half. The continuation, the five closure exclusions, and the converse exhaustion are proved in [lemma D.1](#). The proof avoids any monotonicity assumption on  $\theta \mapsto x$ .

### Noncentral gap comparison

The chord representation now compares noncentral middle branches directly. Let  $N \geq 3$ , put  $M = N + 1$ , and set  $\kappa = N/M$ . Every positive noncentral gap of the  $M$ -problem satisfies

$$j \geq 1, \quad \frac{(j+1)\pi}{N} \leq \frac{\pi}{2}. \quad (4.20)$$

Indeed, such gaps satisfy  $j+1 \leq N/2$ . Hence  $\theta_0 = \theta/\kappa$  belongs to the same-indexed  $N$ -nodal interval, and every trigonometric argument below lies in  $(0, \pi/2]$ . Consider a point of such a branch for the matrix of order  $M - 1$ , with

$$\theta \in \left( \frac{j\pi}{M}, \frac{(j+1)\pi}{M} \right), \quad \frac{(j+1)\pi}{N} \leq \frac{\pi}{2}.$$

First suppose  $\Phi_M(\theta) = \Psi_M(\eta)$ . Put  $\theta_0 = \theta/\kappa$ . Then  $\theta_0$  belongs to the corresponding  $N$ -nodal interval. Concavity of sine gives

$$\frac{\Phi_N(\theta_0)}{\Phi_M(\theta)} = \frac{\mathcal{F}_h(\theta_0)}{\mathcal{F}_h(\theta)} > \kappa. \quad (4.21)$$

Indeed,  $\theta = \kappa\theta_0$  and strict concavity gives  $\sin \theta > \kappa \sin \theta_0$ ; hence

$$\mathcal{F}_h(\theta_0)^2 - \kappa^2 \mathcal{F}_h(\theta)^2 = (1 - \kappa^2) + \mathfrak{s}_h^2 \left( \frac{1}{\sin^2 \theta_0} - \frac{\kappa^2}{\sin^2 \theta} \right) > 0.$$

On the other hand,

$$\frac{\Psi_N(\eta)}{\Psi_M(\eta)} = \frac{\sinh(N\eta)}{\sinh(M\eta)} \leq \kappa, \quad (4.22)$$

with strict inequality for  $\eta > 0$ , by convexity of  $\sinh$  on the positive half-axis. Hence  $\Phi_N(\theta_0) > \Psi_N(\eta)$ . Since  $\Phi_N$  vanishes at the right endpoint  $(j+1)\pi/N$ , its rightmost solution  $\hat{\theta}$  of

$$\Phi_N(\hat{\theta}) = \Psi_N(\eta)$$

exists by the intermediate value theorem and satisfies  $\hat{\theta} > \theta_0 > \theta$ . Strict log-concavity of  $\mathcal{W}_N$  makes it the unique solution on the descending half of that lobe. The same exterior variable is endpoint-side for the  $N$ -problem as well. Indeed, for  $u = \cosh \eta > 1$ ,

$$\frac{d}{du} \log \mathcal{W}_M(u) - \frac{d}{du} \log \mathcal{W}_N(u) = \frac{M \coth(M\eta) - N \coth(N\eta)}{\sinh \eta} > 0.$$

The numerator is positive because, for  $\eta > 0$ ,

$$\frac{d}{dt}(t \coth(t\eta)) = \frac{\sinh(t\eta) \cosh(t\eta) - t\eta}{\sinh^2(t\eta)} > 0.$$

The endpoint-side branch is characterized by a nonpositive logarithmic derivative. The displayed difference therefore makes the  $N$ -derivative strictly negative. The case  $\eta = 0$  follows by continuity. The chord classification above says that the constructed value  $\mathfrak{s}_{\text{hyp}}(\hat{\theta}, \eta)$  is the least singular value at its constructed abscissa. Formula (4.16) and the strict increase of sine on the interval now give

$$\mathfrak{s}_{\text{hyp}}(\hat{\theta}, \eta) > \mathfrak{s}_{\text{hyp}}(\theta, \eta). \quad (4.23)$$

Suppose next that the exterior endpoint is elliptic, so that  $\Phi_M(\theta) = \Phi_M(\phi)$ ,  $0 < \phi < \pi/M$ . (The case  $\phi = 0$  is the limiting hyperbolic case already covered.) Put

$$\theta_0 = \theta/\kappa, \quad \phi_0 = \phi/\kappa.$$

The ratio

$$\mathcal{A}_{N,M}^\Phi(t) = \frac{\Phi_N(t/\kappa)}{\Phi_M(t)} = \frac{\mathcal{F}_h(t/\kappa)}{\mathcal{F}_h(t)}$$

is strictly increasing while  $0 < t < t/\kappa \leq \pi/2$ . Indeed,

$$\frac{d}{dt} \log \mathcal{A}_{N,M}^\Phi(t) = \mathfrak{d}_h(t) - \kappa^{-1} \mathfrak{d}_h(t/\kappa) > 0, \quad \mathfrak{d}_h(t) = \frac{\mathfrak{s}_h^2 \cot t}{\mathfrak{s}_h^2 + \sin^2 t},$$

because

$$t \mathfrak{d}_h(t) = (t \cot t) \frac{\mathfrak{s}_h^2}{\mathfrak{s}_h^2 + \sin^2 t}$$

is strictly decreasing: both positive factors decrease on  $(0, \pi/2)$ , and the second decreases strictly. Since  $\theta > \pi/M > \phi$ , it follows that  $\Phi_N(\theta_0) > \Phi_N(\phi_0)$ . The rightmost solution of

$$\Phi_N(\hat{\theta}) = \Phi_N(\phi_0)$$

therefore has  $\hat{\theta} > \theta_0$ . Its exterior variable is endpoint-side. Indeed, if  $\Lambda_K(t) = (\log \Phi_K)'(t)$ , then on the endpoint-side part of the first outer lobe  $\Lambda_M(\phi) \geq 0$ , and differentiation of  $\Phi_N(t/\kappa) = \mathcal{A}_{N,M}^\Phi(t) \Phi_M(t)$  gives

$$\kappa^{-1} \Lambda_N(t/\kappa) = \Lambda_M(t) + \frac{(\mathcal{A}_{N,M}^\Phi)'(t)}{\mathcal{A}_{N,M}^\Phi(t)} > 0.$$

Because  $\Lambda_K(\phi) = -\sin \phi (\log \mathcal{W}_K)'(\cos \phi)$ , positivity of  $\Lambda_N(\phi_0)$  is exactly the endpoint-side condition for the  $N$ -outer solution. The one-maximum property now fixes the label, and the chord classification says again that  $\mathfrak{s}_{\text{ell}}(\hat{\theta}, \phi_0)$  is the least singular value at the constructed abscissa. Here  $\hat{\theta} > \theta_0 > \theta$  and  $\phi_0 > \phi$ , with all four angles below  $\pi/2$ . Thus both sine factors in the elliptic version of (4.16) have increased, at least one strictly, and hence

$$\mathfrak{s}_{\text{ell}}(\hat{\theta}, \phi_0) > \mathfrak{s}_{\text{ell}}(\theta, \phi). \quad (4.24)$$

Equations (4.23) and (4.24) give the exact conclusion needed from the chord construction: every noncentral branch value for the larger matrix is strictly dominated by a value in the corresponding gap of the smaller matrix. Only the parity-dependent central gaps remain.

*Central-gap interlacing and upper comparison  $d_m < c_m$*

The parity change shifts the central-gap indices, so their comparison is encoded in two alternating central heights.

For  $m \geq 1$ , let

$$c_m = s_{\min}(aI_m + V_m). \quad (4.25)$$

By reflection, the two gaps adjacent to the zero eigenvalue of  $A_{2m+1}$  have the same height; denote it by

$$d_m = \max_{0 \leq x \leq 2r \sin \frac{\pi}{2m+2}} s_{\min}(xI_{2m+1} - A_{2m+1}). \quad (4.26)$$

We claim

$$c_m > d_m > c_{m+1}. \quad (4.27)$$

Put

$$\zeta = \frac{a + a^{-1}}{2}.$$

For every  $j \geq 1$ , there is a unique  $\rho_j > \cos \frac{\pi}{j+1}$  such that

$$aU_j(\rho_j) = U_{j-1}(\rho_j). \quad (4.28)$$

Indeed, the Christoffel–Darboux identity gives, above the largest zero of  $U_j$ ,

$$\frac{d}{dt} \frac{U_{j-1}(t)}{U_j(t)} = -\frac{2 \sum_{k=0}^{j-1} U_k(t)^2}{U_j(t)^2} < 0.$$

The ratio decreases from  $+\infty$  to 0, proving existence and uniqueness. A three-term determinant recurrence gives

$$\det \left( (aI_j + V_j)^* (aI_j + V_j) - (1 + a^2 - 2at)I_j \right) = a^j \left( U_j(t) - a^{-1}U_{j-1}(t) \right). \quad (4.29)$$

Thus the largest root  $\rho_j$  gives the least eigenvalue. Together with

$$U_j(t)^2 + U_{j-1}(t)^2 - 2tU_j(t)U_{j-1}(t) = 1,$$

equations (4.28) and (4.29) give

$$c_j^2 = 1 + a^2 - 2a\rho_j = \frac{1}{U_j(\rho_j)^2}, \quad \zeta - \rho_j = \frac{1}{2U_{j-1}(\rho_j)U_j(\rho_j)}. \quad (4.30)$$

Next we identify  $c_m$  as the exact height of the central gap of  $A_{2m}$ . Let

$$P_{\text{oe}} = [\mathbf{e}_1, \mathbf{e}_3, \dots, \mathbf{e}_{2m-1}, \mathbf{e}_2, \mathbf{e}_4, \dots, \mathbf{e}_{2m}]$$

be the permutation matrix that places odd coordinates first. Then

$$P_{\text{oe}}^T A_{2m} P_{\text{oe}} = \begin{bmatrix} 0 & I_m + aV_m^T \\ aI_m + V_m & 0 \end{bmatrix}.$$

Moreover,

$$(I_m + aV_m^T)^* (I_m + aV_m^T) - J_m (aI_m + V_m)^* (aI_m + V_m) J_m = (1 - a^2) \mathbf{e}_m \mathbf{e}_m^T \geq 0.$$

The singular values of the block off-diagonal matrix are the union of the singular values of its two off-diagonal blocks. The Gram inequality shows that  $s_{\min}(I_m + aV_m^T) \geq$

$s_{\min}(aI_m + V_m) = c_m$ , and hence  $s_{\min}(A_{2m}) = c_m$ . To identify the maximum throughout the central gap, put  $K = 2m + 1$  and use the half variables

$$y = \cos \theta, \quad z = \cosh \eta \quad \text{or} \quad z = \cos \phi.$$

On the positive half of the central middle branch, the preceding chord classification gives

$$\mathcal{W}_K(y) = \mathcal{W}_K(z), \quad x = \frac{2r}{\mathfrak{c}_h} yz, \quad s^2 = \frac{4a}{\mathfrak{c}_h^2} (\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2), \quad (4.31)$$

where  $0 \leq y < \cos \frac{m\pi}{K}$  is in the central lobe and  $z$  is the endpoint-side outer solution. The function  $\mathcal{W}_K$  is even, and strict log-concavity on the central lobe gives

$$\mathcal{W}_K(y) < \mathcal{W}_K(0) = \mathfrak{c}_h \quad (y > 0).$$

Let

$$z_0 = \sqrt{\frac{1 + \rho_m}{2}}.$$

The half-angle identity and (4.28) give

$$U_{2m}(z_0) = U_m(\rho_m) + U_{m-1}(\rho_m) = (1 + a)U_m(\rho_m).$$

Together with (4.30), this yields

$$\mathcal{W}_K(z_0) = \mathfrak{c}_h, \quad 4a(\mathfrak{c}_h^2 - z_0^2) = 2a(\zeta - \rho_m) = c_m^2. \quad (4.32)$$

At  $x = 0$ , lemma 4.2 and the chord-exhaustion argument identify the nearest root with  $c_m$ , so  $z_0$  is the endpoint-side outer solution of level  $\mathfrak{c}_h$ . If  $x \neq 0$ , the level in (4.31) is strictly below  $\mathfrak{c}_h$ , and hence its endpoint-side solution satisfies  $z > z_0$ . Equations (4.31)–(4.32) then give

$$s^2 < 4a(\mathfrak{c}_h^2 - z_0^2) = c_m^2.$$

Reflection gives the same strict bound on the other half of the gap. Therefore the central-gap height equals  $c_m$  and has the unique maximizer  $x = 0$ .

Set

$$\omega_m^e = \cos \frac{\pi}{m+1}, \quad \mathcal{V}_m(t) = \sqrt{(\zeta - t)(1 + t)} U_m(t).$$

For the central specialization, take  $K = 2m + 2$  and set

$$t_{\text{in}} = 2y^2 - 1 = \cos(2\theta), \quad t_{\text{out}} = 2z^2 - 1.$$

On the positive central gap,  $-1 < t_{\text{in}} < -\omega_m^e$ , while the endpoint-side outer variable satisfies  $\omega_m^e < t_{\text{out}} < \zeta$ , where  $\zeta = 2\mathfrak{c}_h^2 - 1$ . The identity  $U_{2m+1}(t) = 2tU_m(2t^2 - 1)$  gives

$$\mathcal{W}_{2m+2}(y)^2 = (\zeta - t_{\text{in}})(1 + t_{\text{in}})U_m(t_{\text{in}})^2, \quad \mathcal{W}_{2m+2}(z)^2 = (\zeta - t_{\text{out}})(1 + t_{\text{out}})U_m(t_{\text{out}})^2.$$

Because  $\zeta + 1 = 2\mathfrak{c}_h^2$ , the formulas  $x = (2r/\mathfrak{c}_h)yz$  and  $s^2 = (4a/\mathfrak{c}_h^2)(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2)$  become the following chord parametrization:

$$\begin{aligned} -1 < t_{\text{in}} < -\omega_m^e, \quad \omega_m^e < t_{\text{out}} < \zeta, \quad \mathcal{V}_m(t_{\text{out}}) &= |\mathcal{V}_m(t_{\text{in}})|, \\ x^2 &= \frac{2a(1 + t_{\text{out}})(1 + t_{\text{in}})}{\zeta + 1}, \quad s^2 = \frac{2a(\zeta - t_{\text{out}})(\zeta - t_{\text{in}})}{\zeta + 1}, \end{aligned} \quad (4.33)$$

where  $t_{\text{out}}$  is the descending, or largest, solution. To see this from (4.15), use

$$\mathfrak{s}_h^2 + \sin^2 \theta = \frac{\zeta - t_{\text{in}}}{2}, \quad \mathfrak{s}_h^2 - \sinh^2 \eta = \frac{\zeta - t_{\text{out}}}{2}.$$

Continuation through  $t_{\text{out}} = 1$  uses the elliptic coordinate  $z = \cos \phi$ .

Let  $\rho = \rho_m$ . The function  $\log \mathcal{V}_m$  is strictly concave on  $(\omega_m^e, \zeta)$ . If  $(\log \mathcal{V}_m)'(\rho) \geq 0$ , then  $\rho$  lies at or before the unique lobe maximum. The descending root lies at or after that maximum, so  $t_{\text{out}} \geq \rho$ . Now suppose  $(\log \mathcal{V}_m)'(\rho) < 0$ . We claim that  $\rho > a$ . If  $\rho \leq a < 1$ , then  $\rho < 1$ , hence  $\rho^2 - 1 \neq 0$ , and the following quotient is well defined:

$$\frac{U'_m(\rho)}{U_m(\rho)} = \frac{m\rho - (m+1)a}{\rho^2 - 1}$$

would give

$$\begin{aligned} (\log \mathcal{V}_m)'(\rho) &= \frac{(2m+2)a + 1 - (2m+1)\rho}{2(1-\rho^2)} - \frac{1}{2(\zeta-\rho)} \\ &\geq \frac{1}{2(1-\rho)} - \frac{1}{2(\zeta-\rho)} \geq 0, \end{aligned}$$

which contradicts the assumed negative derivative. Hence  $\rho > a$ . Writing  $t_{\text{in}} = -\cos \phi_c$ ,  $0 < \phi_c < \pi/(m+1)$ , gives

$$\begin{aligned} |\mathcal{V}_m(t_{\text{in}})|^2 &= \frac{(\zeta + \cos \phi_c) \sin^2((m+1)\phi_c)}{1 + \cos \phi_c} \\ &< \frac{\zeta + \omega_m^e}{1 + \omega_m^e} < \frac{1+a}{2a} < \frac{1+\rho}{2a} = \mathcal{V}_m(\rho)^2. \end{aligned}$$

Now  $\rho$  is on the descending side and  $|\mathcal{V}_m(t_{\text{in}})| < \mathcal{V}_m(\rho)$ , so the unique descending root again satisfies  $t_{\text{out}} > \rho$ . In either case,  $t_{\text{out}} \geq \rho$ , and  $t_{\text{in}} > -1$  makes the following inequality strict:

$$s^2 < 2a(\zeta - \rho) = c_m^2.$$

This holds along the whole central sheet, so  $d_m < c_m$ . Thus the odd-order central height lies strictly below the preceding even-order central height. To complete the alternating comparison, it remains to place it strictly above the next even-order height.

*Lower comparison  $d_m > c_{m+1}$ : decisive scalar estimate*

Put  $L = m+1 \geq 2$ , and abbreviate

$$\rho = \rho_L, \quad u_- = U_{L-1}(\rho), \quad u_+ = U_L(\rho), \quad \omega_L = \cos \frac{\pi}{2L}, \quad \Delta_\omega = 1 - \omega_L,$$

$$\mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+, \quad \xi = \rho - \frac{\Delta_\omega}{\mathfrak{D}_L}. \quad (4.34)$$

Thus  $a = u_-/u_+$  and  $c_L = 1/u_+$ .

The required inequality is

$$(1 + \omega_L)(1 + \xi)^2 U_{L-1}(\xi)^2 < (\rho + \omega_L) \mathfrak{D}_L. \quad (4.35)$$

The proof of this strict scalar estimate is isolated in [appendix E](#). It combines the Chebyshev identity with strict log-concavity on  $[\xi, \rho]$ , and then treats the three exhaustive cases  $\rho > 1$ ,  $\rho = 1$ , and  $\rho < 1$ . In the last case, explicit finite- $L$  bounds close the only delicate endpoint estimate. This proves (4.35) for every  $L \geq 2$  by exact finite-dimensional estimates.

The remaining task is to certify that the signed root selected by this estimate is the least singular value on the actual central gap.

*Certification of the least singular value*

Since  $a > 0$ ,  $\Delta_\omega > 0$ ,  $\rho + \omega_L > 0$ , and  $\zeta + \omega_L > 0$ , let  $x_* > 0$  be the positive square root determined by

$$x_*^2 = \frac{2a\Delta_\omega(\rho + \omega_L)}{\zeta + \omega_L}, \quad c = c_L. \quad (4.36)$$

The algebraic certification is given in [appendix F](#). Substitution of the distinct folded pair  $u = \xi$ ,  $v = -\omega_L$  into (4.11), followed by (4.35), proves

$$q_{2L-1}(x_*, c)q_{2L-1}(x_*, -c) > 0. \quad (4.37)$$

At  $x = 0$ , the second-smallest singular value of  $A_{2L-1}$  is

$$s_{L,2}^{(0)} = \sqrt{1 + a^2 - 2a \cos \frac{\pi}{L}}.$$

The same appendix proves the strict separation

$$x_* + c < 2\sqrt{a} \sin \frac{\pi}{2L} \leq s_{L,2}^{(0)}. \quad (4.38)$$

The product parity and this second-singular-value separation now identify the relevant signed root independently of any prior branch choice.

Let  $\mu_1, \dots, \mu_{2L-1}$  be the eigenvalues of the real symmetric matrix

$$B_{2L-1}(x_*) = (x_*I_{2L-1} - A_{2L-1})J_{2L-1}.$$

Right multiplication by  $J_{2L-1}$  preserves singular values, so the numbers  $|\mu_i|$  are precisely the singular values of  $x_*I_{2L-1} - A_{2L-1}$ . Moreover,

$$q_{2L-1}(x_*, c)q_{2L-1}(x_*, -c) = \prod_{i=1}^{2L-1} (\mu_i^2 - c^2).$$

Positivity of the product in (4.37) implies that the number of indices with  $|\mu_i| < c$  is even. On the other hand, Weyl's singular-value perturbation inequality and (4.38) give, for the second-smallest singular value  $s_{\text{next}}$ ,

$$s_{\text{next}}(x_*I_{2L-1} - A_{2L-1}) \geq s_{L,2}^{(0)} - x_* > c,$$

so at most one index can satisfy  $|\mu_i| < c$ . Parity therefore gives  $|\mu_i| \geq c$  for every  $i$ , and strict positivity of the product rules out equality. Hence  $s_{\min}(x_*I_{2L-1} - A_{2L-1}) > c_L$ . Since  $0 < x_* < 2r \sin \frac{\pi}{2L}$ , the point lies strictly in the positive central gap, and  $d_{L-1} > c_L$ . This completes the proof of (4.27). Together with the upper comparison, this proves the strict alternating chain  $c_m > d_m > c_{m+1}$ , exactly resolving the central-gap obstruction left by the noncentral chord comparison.

*Assembly of the global barrier inequality*

Reflection and the central interlacing chain now assemble the local comparisons into the global barrier inequality. Put  $M = N + 1$ , so the larger and smaller matrices have orders  $M - 1$  and  $N - 1$ , respectively. If  $\Sigma = \text{diag}(1, -1, 1, -1, \dots)$ , then  $\Sigma A_n \Sigma = -A_n$ , so reflection reduces every noncentral comparison to a positive gap. By the interiority observation following the definition of the gap heights, choose  $x_j \in G_{M-1,j}^\circ$  at which

its height is attained. The hyperbolic or elliptic construction above produces a point  $\hat{x}_j \in G_{N-1,j}$ , and chord exhaustion gives

$$\gamma_{N-1,j} \geq g_{N-1}(\hat{x}_j) > g_{M-1}(x_j) = \gamma_{M-1,j}.$$

Reflection gives the same inequality on the corresponding reflected gap.

If  $M = 2\ell_* + 1$ , the remaining single central height of the larger, even-order matrix is  $c_{\ell_*}$ , whereas each central height of the smaller odd-order matrix is  $d_{\ell_*-1} > c_{\ell_*}$ . If  $M = 2\ell_* + 2$ , the two central heights of the larger odd-order matrix equal  $d_{\ell_*} < c_{\ell_*}$ , where  $c_{\ell_*}$  is the single central height of the smaller even-order matrix. Thus every height of the larger problem is strictly smaller than at least one height of the smaller problem. Choose  $j_*$  with  $\gamma_{M-1,j_*} = \max_j \gamma_{M-1,j}$ . Then

$$\gamma_{M-1} = \gamma_{M-1,j_*} < \max_k \gamma_{N-1,k} = \gamma_{N-1}.$$

Since  $M - 1 = n + 1$  and  $N - 1 = n$ , this proves  $\gamma_{n+1} < \gamma_n$ .  $\square$

## 5 Topology, quantitative bounds, and the transition scale

The first topological ingredient is the standard component property for arbitrary matrices [16]. In the present setting it reduces the planar topology to the real spectral interval.

**Lemma 5.1.** *Every connected component of  $\sigma_\varepsilon(A_n)$  contains an eigenvalue of  $A_n$ .*

*Proof.* The estimate

$$s_{\min}(zI_n - A_n) \geq |z| - \|A_n\|_2$$

shows that the pseudospectrum is bounded. Let  $\mathcal{C}$  be one of its components. Any eigenvalue  $\lambda \in \overline{\mathcal{C}}$  has, by continuity of the least singular value, a ball contained in the pseudospectrum. That ball meets  $\mathcal{C}$ , so maximality of connected components places  $\lambda$  in  $\mathcal{C}$ .

To prove that  $\mathcal{C}$  contains an eigenvalue, assume instead that it lies entirely in the resolvent set. The preceding observation then places  $\overline{\mathcal{C}}$  in the resolvent set as well. On a neighbourhood of  $\overline{\mathcal{C}}$ , the function  $\mathbf{u}(z) = \|(zI_n - A_n)^{-1}\|_2$  is continuous and subharmonic; see [3]. Moreover,  $\mathbf{u} \leq \varepsilon^{-1}$  on  $\partial\mathcal{C}$ : a boundary point with  $\mathbf{u} > \varepsilon^{-1}$  would have a neighbourhood in the pseudospectrum that meets  $\mathcal{C}$ , and hence would belong to  $\mathcal{C}$ . The maximum principle gives  $\mathbf{u} \leq \varepsilon^{-1}$  throughout  $\mathcal{C}$ , whereas the definition of the pseudospectrum gives  $\mathbf{u} > \varepsilon^{-1}$  there, a contradiction.  $\square$

The diagonal gauge introduced in the introduction gives

$$A_n = \mathcal{G}_n(r) [r(V_n + V_n^T)] \mathcal{G}_n(r)^{-1},$$

and hence

$$\sigma(A_n) = \left\{ 2r \cos \frac{k\pi}{n+1} : 1 \leq k \leq n \right\}. \quad (5.1)$$

Put

$$\Omega = \sigma_\varepsilon(A_n), \quad \Omega_{\mathbb{R}} = \Omega \cap \mathbb{R} = \{x \in \mathbb{R} : g_n(x) < \varepsilon\}.$$

Because  $A_n$  is real, the least singular value is invariant under complex conjugation. Lemma 3.1 therefore implies that

$$\mathcal{H}_t(x + iy) = x + i(1-t)y, \quad 0 \leq t \leq 1,$$

maps  $\Omega$  into itself. If  $\mathcal{C}$  is a component of  $\Omega$ , the path  $t \mapsto \mathcal{H}_t(z)$  starts in  $\mathcal{C}$  and hence remains there. Thus  $\mathcal{H}_t$  retracts  $\mathcal{C}$  onto  $\mathcal{C} \cap \mathbb{R}$ .

By Lemma 5.1, this real intersection is nonempty. As the continuous image  $\mathcal{H}_1(\mathcal{C})$ , it is connected and lies in an interval component  $\mathcal{K}$  of  $\Omega_{\mathbb{R}}$ . Because  $\mathcal{K}$  is connected and meets  $\mathcal{C}$ , maximality gives  $\mathcal{K} \subset \mathcal{C}$ . Hence  $\mathcal{C} \cap \mathbb{R} = \mathcal{K}$ , so the components of  $\Omega$  correspond bijectively to the interval components of  $\Omega_{\mathbb{R}}$ . The retraction therefore makes every component contractible.

If  $\Omega$  is connected, its real intersection is an interval containing every eigenvalue, and therefore contains their convex hull  $I_n^{\text{sp}}$ . Conversely, if  $I_n^{\text{sp}} \subset \Omega$ , it lies in a component containing every eigenvalue. Lemma 5.1 assigns an eigenvalue to every component, so this component is all of  $\Omega$ . Compactness of  $I_n^{\text{sp}}$  now gives

$$\Omega \text{ is connected} \iff I_n^{\text{sp}} \subset \Omega \iff \max_{x \in I_n^{\text{sp}}} g_n(x) < \varepsilon \iff \gamma_n < \varepsilon. \quad (5.2)$$

The strict inequality reflects the open definition of the pseudospectrum. At  $\varepsilon = \gamma_n$ , a maximizing barrier point remains excluded.

The connectedness criterion is now exact; the next proposition determines the scale of  $\gamma_n$ . Put  $N = n + 1$  and recall  $\mathcal{L}_n(a)$  and  $\mathcal{U}_n(a)$  from (2.2)–(2.3).

**Proposition 5.2** (Sharp-order gap bounds). *For every  $n \geq 2$ ,*

$$\mathcal{L}_n(a) \leq \gamma_n \leq \mathcal{U}_n(a). \quad (5.3)$$

*In particular, both bounds have order  $(n + 1)a^{n/2}$ , with constants depending only on  $a$ .*

*Proof.* For  $x = 2r \cos \theta \in I_n^{\text{sp}}$ , set

$$\mathbf{v} = (v_j)_{j=1}^n, \quad v_j = r^{j-1} \sin(j\theta), \quad 1 \leq j \leq n.$$

The residual  $(xI_n - A_n)\mathbf{v}$  is supported in its last coordinate, where it equals  $r^n \sin(N\theta)$ . Thus

$$g_n(2r \cos \theta) \leq \frac{r^n |\sin(N\theta)|}{\left(\sum_{j=1}^n r^{2j-2} \sin^2(j\theta)\right)^{1/2}} \leq r^n \frac{|\sin(N\theta)|}{\sin \theta}.$$

Since  $\pi/N \leq \theta \leq (N - 1)\pi/N$ , this proves the upper bound.

For the lower bound, put  $p_\theta = re^{i\theta}$ ,  $q_\theta = re^{-i\theta}$ , so that  $p_\theta + q_\theta = x$  and  $p_\theta q_\theta = a$ . For  $\mathbf{u} = (u_j)_{j=1}^n \in \mathbb{C}^n$ , put  $u_0 = u_{n+1} = 0$ , and define maps from  $\mathbb{C}^n$  to  $\mathbb{C}^{n+1}$  by

$$(\mathbf{C}_\theta \mathbf{u})_j = p_\theta u_j - u_{j+1}, \quad (\mathbf{D}_\theta \mathbf{u})_j = u_j - \overline{q_\theta} u_{j+1}, \quad 0 \leq j \leq n.$$

For  $1 \leq k \leq n$ , direct multiplication gives

$$(\mathbf{D}_\theta^* \mathbf{C}_\theta \mathbf{u})_k = (\mathbf{C}_\theta \mathbf{u})_k - q_\theta (\mathbf{C}_\theta \mathbf{u})_{k-1} = (p_\theta + q_\theta) u_k - u_{k+1} - p_\theta q_\theta u_{k-1}.$$

Hence

$$xI_n - A_n = \mathbf{D}_\theta^* \mathbf{C}_\theta. \quad (5.4)$$

The two Gram matrices are diagonally unitarily similar to the tridiagonal Toeplitz matrix with diagonal  $1 + r^2$  and off-diagonal  $-r$ . Its eigenvalues are  $1 + r^2 - 2r \cos \frac{k\pi}{N}$ ,  $1 \leq k \leq n$ . Consequently, for every  $\mathbf{u} \in \mathbb{C}^n$ ,

$$\begin{aligned} \|\mathbf{C}_\theta \mathbf{u}\|_2 &\geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2} \|\mathbf{u}\|_2, \\ \|\mathbf{D}_\theta \mathbf{u}\|_2 &\geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2} \|\mathbf{u}\|_2. \end{aligned} \quad (5.5)$$

Put

$$\mathcal{S}_N = \sum_{j=0}^n r^{2j} = \frac{1 - r^{2N}}{1 - r^2}.$$

Unit normal vectors to the two hyperplane ranges are

$$\mathbf{u}_C = \mathcal{S}_N^{-1/2}(\overline{p}_\theta^n, \overline{p}_\theta^{n-1}, \dots, 1)^T, \quad \mathbf{u}_D = \mathcal{S}_N^{-1/2}(1, q_\theta, \dots, q_\theta^n)^T.$$

In the polar decompositions

$$\mathbf{C}_\theta = Q_C(\mathbf{C}_\theta^* \mathbf{C}_\theta)^{1/2}, \quad \mathbf{D}_\theta = Q_D(\mathbf{D}_\theta^* \mathbf{D}_\theta)^{1/2},$$

the matrices  $Q_C, Q_D$  are isometries onto these ranges. Two hyperplanes in  $\mathbb{C}^{n+1}$  intersect in dimension  $n-1$ , so the cross-Gram matrix  $Q_D^* Q_C$  has  $n-1$  singular values equal to 1; its final singular value is  $|\langle \mathbf{u}_D, \mathbf{u}_C \rangle|$ , the cosine of the nontrivial principal angle. The geometric sum gives

$$\begin{aligned} |\langle \mathbf{u}_D, \mathbf{u}_C \rangle| &= \frac{r^n}{\mathcal{S}_N} \left| \sum_{j=0}^{N-1} e^{i(2j-(N-1))\theta} \right| \\ &= \frac{r^n(1-r^2)}{1-r^{2N}} \left| \frac{\sin(N\theta)}{\sin\theta} \right| =: \chi_N(\theta). \end{aligned} \quad (5.6)$$

Finally,

$$xI_n - A_n = (\mathbf{D}_\theta^* \mathbf{D}_\theta)^{1/2} Q_D^* Q_C (\mathbf{C}_\theta^* \mathbf{C}_\theta)^{1/2}.$$

Using  $s_{\min}(XYZ) \geq s_{\min}(X)s_{\min}(Y)s_{\min}(Z)$  and (5.5) gives

$$g_n(2r \cos \theta) \geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right) \chi_N(\theta).$$

Putting  $\theta = 3\pi/(2N)$ , which lies in the first spectral gap, proves the lower bound.

Finally,

$$\mathcal{U}_n(a) \leq \frac{1}{2}(n+1)r^n,$$

while

$$\mathcal{L}_n(a) \geq C_{\text{lb}}(a)(n+1)r^n, \quad C_{\text{lb}}(a) = \frac{2(1-r)^2(1-r^2)}{3\pi}. \quad (5.7)$$

These follow from  $\sin \frac{\pi}{N} \geq 2/N$ ,  $1 + r^2 - 2r \cos \frac{\pi}{N} \geq (1-r)^2$ , and  $\sin \frac{3\pi}{2N} \leq 3\pi/(2N)$ . This proves the asserted sharp order.  $\square$

*Proof of Theorem 2.1.* By Proposition 5.2,  $\gamma_n \leq \mathcal{U}_n(a) \rightarrow 0$ . Lemma 4.1 says that  $\gamma_n$  is strictly decreasing. Equation (5.2) therefore shows that, once the pseudospectrum is connected, it stays connected in every larger dimension. This proves  $N_e = N_f$ . Contractibility of every component was proved immediately before (5.2).

If  $\mathcal{L}_n(a) \geq \varepsilon$ , then  $\gamma_n \geq \varepsilon$ , so the pseudospectrum in dimension  $n$  is disconnected. If  $\mathcal{U}_n(a) < \varepsilon$ , then  $\gamma_n < \varepsilon$ , so it is connected. Strict decrease of  $\gamma_n$  now gives (2.4).

A closed-form inversion is available for the comparison envelope that brackets  $N_c$ . Put

$$\Theta_n = (n+1)r^n, \quad \mathcal{T}_r(t) = \min\{N \geq 2 : \Theta_m < t \text{ for every } m \geq N\}.$$

Equations (5.7) and  $\mathcal{U}_n(a) \leq \Theta_n/2$  give

$$\boxed{\mathcal{T}_r(\varepsilon/C_{\text{lb}}(a)) \leq N_c(a, \varepsilon) \leq \mathcal{T}_r(2\varepsilon)}. \quad (5.8)$$

Indeed, (5.2) gives

$$N_c(a, \varepsilon) = \min\{N \geq 2 : \gamma_m < \varepsilon \text{ for every } m \geq N\}.$$

The lower estimate  $C_{\text{lb}}(a)\Theta_m \leq \gamma_m$  forces  $\Theta_m < \varepsilon/C_{\text{lb}}(a)$  throughout the tail beginning at  $N_c$ . Conversely,  $\Theta_m < 2\varepsilon$  for every  $m \geq \mathcal{T}_r(2\varepsilon)$ , together with  $\gamma_m \leq \Theta_m/2$ , forces connectedness throughout that tail. Thus the polynomial factor  $n+1$ , as well as the exponential factor  $a^{n/2}$ , is sharp on both sides.

*Lambert-W inversion of the envelope threshold*

Let

$$\beta = -\log r = \frac{1}{2} \log(1/a).$$

For  $0 < t < 3r^2 = \Theta_2$ , let  $x_t > 2$  be the solution on the decreasing branch of  $(x+1)e^{-\beta x} = t$ . This branch exists because

$$0 < \beta r t < 3\beta r^3 \leq e^{-1};$$

the last inequality is  $ue^{-u} \leq e^{-1}$  with  $u = 3\beta$ . Moreover,  $t < \Theta_2$  places the descending root strictly beyond  $x = 2$ . Putting  $y_t = x_t + 1$ , the equation becomes  $(-\beta y_t)e^{-\beta y_t} = -\beta r t$ , whence  $y_t = -\beta^{-1}W_{-1}(-\beta r t)$ . Therefore

$$\mathcal{T}_r(t) = \left\lfloor -\frac{1}{\beta} W_{-1}(-\beta r t) \right\rfloor, \quad (5.9)$$

where  $W_{-1}$  is the lower real branch of the Lambert  $W$ -function. Indeed, the continuous function  $(x+1)r^x$  is unimodal and, after its descending root, is strictly below  $t$ . Hence the integer tail condition is  $N > x_t$ ; its least solution is  $\lfloor x_t \rfloor + 1 = \lfloor x_t + 1 \rfloor$ , including when  $x_t$  is an integer.

The standard lower-branch expansion [26]

$$-W_{-1}(-u) = \log(1/u) + \log\log(1/u) + O(1), \quad u \downarrow 0,$$

combined with (5.8) proves, for each fixed  $a \in (0, 1)$ , as  $\varepsilon \downarrow 0$ ,

$$N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log\log(1/\varepsilon)) + O_a(1). \quad (5.10)$$

The remainder is pointwise for fixed  $a$ ; uniform control as  $a \uparrow 1$  lies outside this statement.

*Exact two-site endpoint*

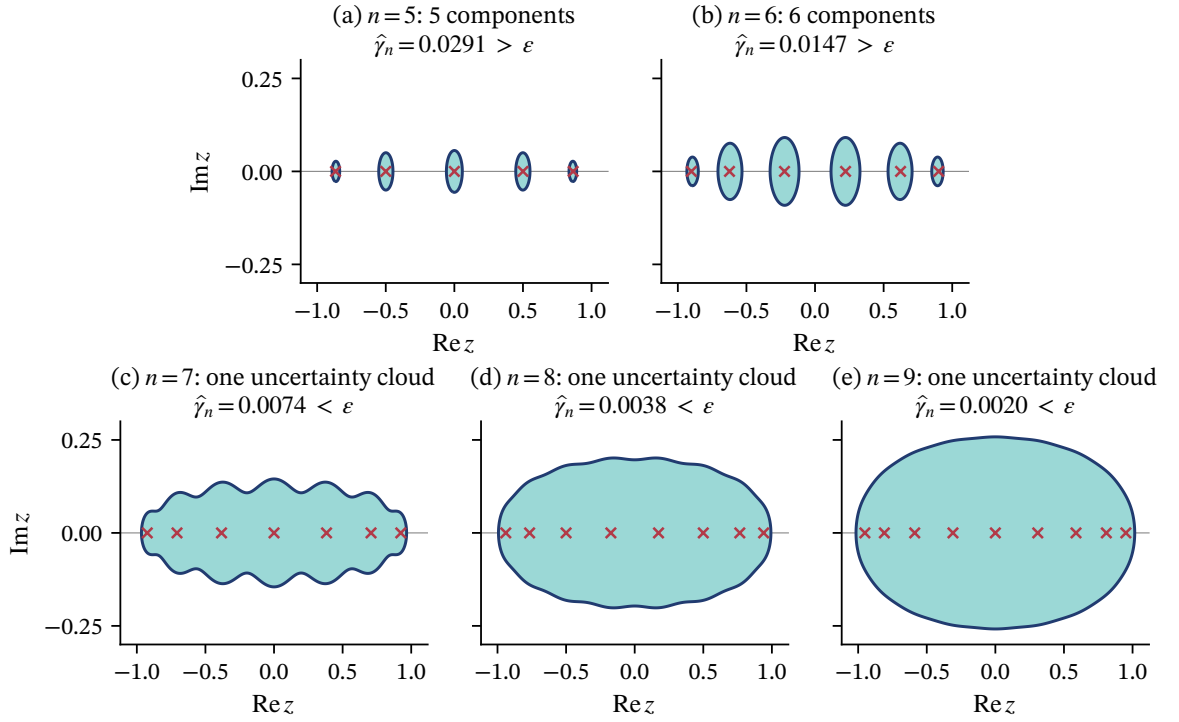
For  $|x| \leq r$ ,

$$\det((xI_2 - A_2)^*(xI_2 - A_2) - a^2 I_2) = x^2(x^2 - 2a - 2a^2) \leq 0.$$

The matrix inside the determinant is Hermitian, and the nonpositive product of its two eigenvalues forces its least eigenvalue to be nonpositive. Thus  $s_{\min}(xI_2 - A_2) \leq a$ . At  $x = 0$ , the singular values of  $-A_2$  are 1 and  $a$ , so equality is attained and  $\gamma_2 = a$ . Hence, by (5.2),

$$N_c(a, \varepsilon) = 2 \iff \varepsilon > a. \quad (5.11)$$

This proves all assertions.  $\square$



**Figure 1.** Numerical illustration of the Euclidean pseudospectral merger for  $A_n(1/4)$  at  $\varepsilon = 10^{-2}$ . Shading shows the numerically sampled strict pseudospectrum and crosses mark the exact open-boundary eigenvalues. Separate gapwise checks support the component counts for  $n = 5, 6$ , while global checks place the illustrated one-cloud transition at  $n = 7$ . The shaded sets and component counts are numerical illustrations; the theorem supplies the connectedness criterion and its persistence.

## 6 Numerical illustrations

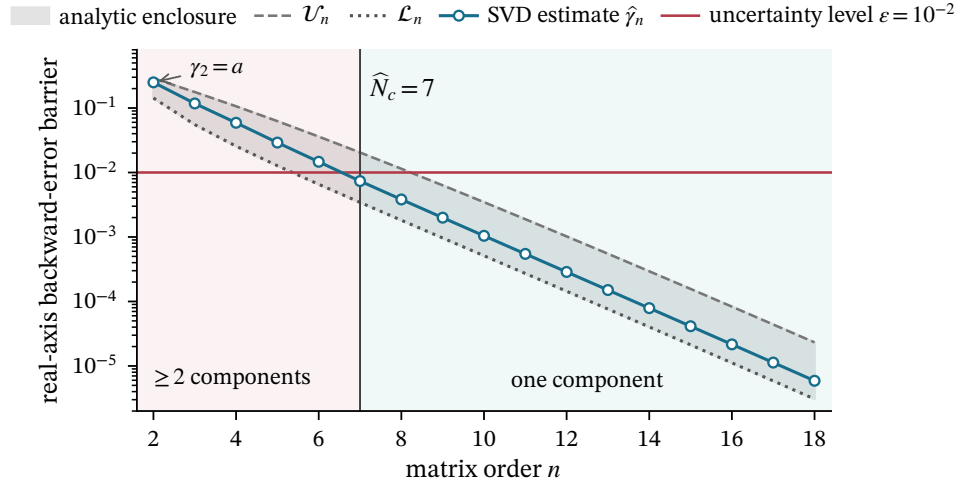
Figures 1 to 3 show three complementary aspects of the finite-size transition: merger of the pseudospectral components, decay of the real-axis backward-error barrier with chain length, and the imaginary-gauge origin of that decay. The figures are numerical illustrations; every theorem and bound is analytic.

Figure 1 shows the final merger at fixed uncertainty. For shorter chains, individual open-boundary eigenvalues retain separate pseudospectral components; at the threshold size these components form one connected uncertainty cloud.

For the plotted samples, the separate gapwise Lipschitz lower checks put every real-gap maximum for  $n = 5, 6$  above  $\varepsilon$ , while the global upper checks put  $\gamma_n$  below it for  $n = 7, 8, 9$ . Thus the first two panels are mode-resolved and the remaining three form a single cloud in the numerical illustration. For the open Hatano–Nelson example, this illustrates the  $\tilde{N}$  cloud-merger crossover reported numerically by Kiorpelidis and Makris [24]; the theorem here identifies the exact connectedness threshold of the full Euclidean pseudospectrum. Once the true pseudospectrum is connected, every vertical section is an interval and the real trace contains the full open-boundary spectral segment.

Figure 2 illustrates the complementary roles of the two analytic results. Strict barrier comparison gives a unique crossing at fixed  $\varepsilon$ , while the upper and lower envelopes establish its sharp exponential-polynomial scale independently of numerical identification of the maximizing gap.

Figure 3 places the connectedness criterion and its nonnormal mechanism side by side. The left panels show the final real-axis barrier crossing the fixed uncertainty level; the right panel shows the mode envelope generated by the same imaginary gauge that sets the exponentially small barrier scale. Reversing the site convention reverses



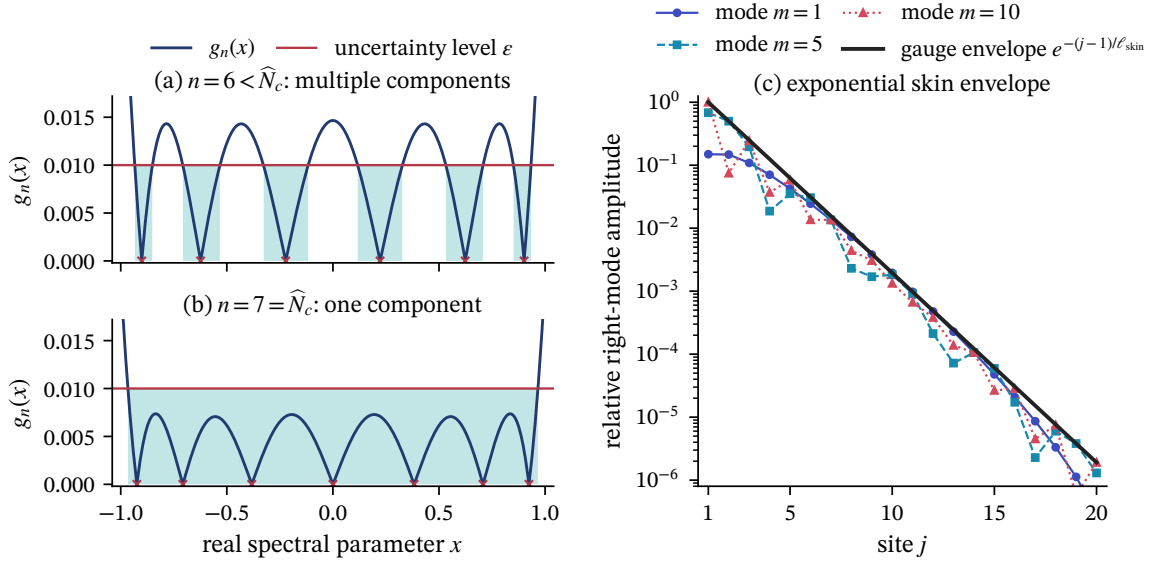
**Figure 2.** Real-axis backward-error barriers for  $a = 1/4$ . The grey band, bounded by the dotted lower curve  $\mathcal{L}_n(a)$  and dashed upper curve  $\mathcal{U}_n(a)$ , is the rigorous enclosure of  $\gamma_n$  and proves the scale  $(n+1)a^{n/2}$ . Blue circles joined by a solid line are floating-point estimates; the red horizontal line is the fixed uncertainty  $\varepsilon = 10^{-2}$ , and the black vertical line marks the numerical crossing  $\hat{N}_c = 7$ . The theorem gives the analytic bracket  $6 \leq N_c \leq 9$ ; the proof uses the analytic bounds.

the localization edge while leaving the connectedness criterion invariant.

For completeness, the numerical classification uses a rigorous sampling implication. Let  $M_{\Delta x}$  be the largest exact least singular value sampled over grids covering every closed gap, each with maximum spacing  $\Delta x$ . The 1-Lipschitz property of  $x \mapsto s_{\min}(xI_n - A_n)$  gives

$$M_{\Delta x} \leq \gamma_n \leq M_{\Delta x} + \frac{\Delta x}{2}.$$

The same argument applies to each individual gap. The plotted inputs use ordinary floating-point arithmetic; interval certification lies beyond these computations, so the figures serve as numerical illustrations.



**Figure 3.** Backward-error barrier and skin-mode envelope for  $a = 1/4$  and  $\varepsilon = 10^{-2}$ . Panels (a) and (b) show numerical samples of  $g_n(x)$ , the uncertainty level, and the resulting real-axis pseudospectral trace for  $n = 6$  and  $n = 7$ . Panel (c) evaluates three right-eigenmode profiles from the closed formula at  $n = 20$ ; the black line is their common imaginary-gauge envelope, whose amplitude decay length is  $\ell_{\text{skin}} = 2/\log(1/a)$ . The numerical evaluations are illustrative; the proof is analytic.

## 7 Physical interpretation: spectral resolution and skin depth

The physical content of the theorem is that skin localization and the finite-size loss of worst-case spectral resolution share one controlling scale. The imaginary gauge fixes both the exponential mode envelope and the exponentially collapsing backward-error barrier; strict barrier comparison turns this scale relation into a unique system-size transition. We now restore the hopping amplitudes and physical units.

To connect the dimensionless matrix to the usual second-quantized notation, consider

$$\widehat{H}_{\text{OBC}} = - \sum_{j=1}^{n-1} \left( t_L \hat{c}_j^\dagger \hat{c}_{j+1} + t_R \hat{c}_{j+1}^\dagger \hat{c}_j \right), \quad t_L > t_R > 0.$$

Its one-particle matrix is  $-t_L A_n(a)$ ,  $a = t_R/t_L$ . The conventional minus sign is immaterial here because  $\Sigma A_n(a) \Sigma = -A_n(a)$  for  $\Sigma = \text{diag}(1, -1, \dots, (-1)^{n-1})$ . Put

$$t_0 = \sqrt{t_L t_R}, \quad h = \frac{1}{2} \log(t_L/t_R) = \frac{1}{2} \log(1/a).$$

For the unitarily equivalent positive-sign representative  $H_n = t_L A_n(a)$ , multiplying the dimensionless gauge relation by  $t_L$  gives the eigenvalues, unnormalized right eigenvectors  $\psi_m^R$  of  $H_n$ , and unnormalized column eigenvectors  $\psi_m^L$  of  $H_n^\dagger$ :

$$\begin{aligned} E_m &= 2t_0 \cos \frac{m\pi}{n+1}, \\ \psi_m^R(j) &= e^{-h(j-1)} \sin \frac{j m \pi}{n+1}, \\ \psi_m^L(j) &= e^{h(j-1)} \sin \frac{j m \pi}{n+1}, \quad 1 \leq j, m \leq n. \end{aligned}$$

With this normalization,  $(\psi_m^L)^\dagger \psi_{m'}^R = (n+1) \delta_{mm'}/2$ . These mode profiles are closely related to the original Hatano–Nelson eigenfunction analysis [27]. Thus the amplitude skin depth of the right eigenmodes is

$$\ell_{\text{skin}} = h^{-1} = \frac{2}{\log(1/a)},$$

the amplitude e-folding length. The intensity or probability envelope has e-folding length  $(2h)^{-1} = \ell_{\text{skin}}/2$ , while the exponential gauge factors cancel in the corresponding biorthogonal left-right product. The gauge condition number is  $\kappa_2(\mathcal{G}_n(e^{-h})) = e^{h(n-1)} = a^{-(n-1)/2}$ .

### 7.1 Why a finite threshold exists

The least singular value admits the backward-error formula

$$s_{\min}(zI_n - A_n) = \min\{\|\Delta A\|_2 : z \in \sigma(A_n + \Delta A)\}.$$

Accordingly,  $\gamma_n$  is the largest, over the real interval between the extreme open-boundary eigenvalues, of the minimum perturbation required to place an eigenvalue at a prescribed point. Hence it is exactly the final barrier that the uncertainty level must exceed for connectedness.

For the dimensional positive-sign Hamiltonian  $H_n = t_L A_n(a)$ , define the dimensional barrier

$$\varepsilon_*(n) = t_L \gamma_n.$$

Then the exact physical form of the connectedness criterion is

$$\sigma_{\varepsilon_{\text{phys}}}(H_n) \text{ is connected} \iff \varepsilon_{\text{phys}} > \varepsilon_*(n).$$

Because  $\gamma_n$  decreases strictly to zero, a fixed uncertainty crosses the barrier sequence at one site count, after which the cloud remains connected. Equivalently, if  $E_{\min}$  and  $E_{\max}$  are the extreme eigenvalues of  $H_n$ , then

$$\varepsilon_*(n)^{-1} = \min_{x \in [E_{\min}, E_{\max}]} \|(xI_n - H_n)^{-1}\|_2,$$

with the resolvent norm interpreted as infinite at an eigenvalue. At each  $z$ , this norm is the pointwise worst-case frequency-domain amplification from forcing to response; for  $H_n$  it has physical units of inverse energy. Thus  $\varepsilon_*(n)^{-1}$  is the minimum pointwise amplification over the spectral interval, while  $\gamma_n^{-1}$  is the corresponding dimensionless minimum for  $A_n$ . At and beyond  $N_c$ , this minimum exceeds  $\varepsilon_{\text{phys}}^{-1}$ .

The exponential collapse of the barrier is the direct link to Hatano–Nelson skin physics. From [Proposition 5.2](#),

$$\varepsilon_*(n) \asymp_a t_L(n+1)e^{-hn} = t_0(n+1)e^{-h(n-1)} = t_0(n+1)e^{-(n-1)/\ell_{\text{skin}}}.$$

The nonunitary imaginary gauge becomes exponentially ill-conditioned across the chain, and right and left skin modes accumulate at opposite edges. Accordingly,

$$\gamma_n^{-1} \asymp_a \frac{e^{hn}}{n+1},$$

so gauge conditioning, skin localization, and worst-case spectral sensitivity share the exponential rate  $h$ , with the barrier carrying the additional sharp polynomial factor. The skin effect and the merger transition are distinct phenomena controlled by the same imaginary gauge.

Let  $\delta = \varepsilon_{\text{phys}}/t_L$ . Balancing the fixed uncertainty level with the barrier scale above gives, as  $\delta \downarrow 0$ ,

$$N_c(a, \delta) = \ell_{\text{skin}} (\log(1/\delta) + \log \log(1/\delta)) + O_a(1).$$

Here  $N_c$  counts sites. For lattice spacing  $d$ , the physical end-to-end length is approximately  $(N_c - 1)d$ ; the term spectral-resolution length therefore refers to  $N_c$  measured in lattice-spacing units. In amplitude skin-depth units,

$$\frac{N_c}{\ell_{\text{skin}}} = \log(1/\delta) + \log\log(1/\delta) + O_a(1),$$

which is the threshold expressed in amplitude-skin-depth units. Improving the resolution by a factor  $q$  increases the leading resolvable site count by  $\ell_{\text{skin}} \log q$ . At fixed  $\delta = \varepsilon_{\text{phys}}/t_L$ , stronger nonreciprocity shortens  $\ell_{\text{skin}}$  and hence lowers the leading small- $\delta$  threshold. This conclusion concerns the stated small- $\delta$  asymptotic; finite-parameter monotonicity in  $a$  lies outside the theorem.

The exact dimer provides a useful endpoint check. Since  $\gamma_2 = a$ ,

$$\varepsilon_*(2) = t_L a = t_R, \quad N_c(a, \varepsilon_{\text{phys}}/t_L) = 2 \quad \Longleftrightarrow \quad \varepsilon_{\text{phys}} > t_R.$$

Indeed, for the positive-sign dimer,

$$H_2 = \begin{bmatrix} 0 & t_L \\ t_R & 0 \end{bmatrix}, \quad \Delta H = -t_R \mathbf{e}_2 \mathbf{e}_1^T, \quad H_2 + \Delta H = \begin{bmatrix} 0 & t_L \\ 0 & 0 \end{bmatrix}.$$

This norm- $t_R$  rank-one perturbation cancels the weaker directed hopping and produces a defective double eigenvalue at zero. It realizes the exact endpoint  $\gamma_2 = a$ : when  $\varepsilon_{\text{phys}} > t_R$ , component-based mode resolution is lost already for the dimer. At equality, the perturbation lies on the boundary excluded by the strict convention  $\|\Delta H\|_2 < \varepsilon_{\text{phys}}$ , so the component closures touch while the strict pseudospectrum remains disconnected; indeed,  $N_c(a, a) = 3$ . The rank-one hopping perturbation is an illustrative structured realization. Assessing whether generic onsite or hopping disorder realizes this extremal perturbation requires a structured-disorder analysis.

Kiorpelidis and Makris [24] introduced  $\widehat{N}$  for the observed fixed- $\varepsilon$  crossover from eigenvalue-centred clouds to a single cloud and related it to the end of exponential growth of the pseudospectral abscissa. For the clean open Hatano–Nelson chain,  $N_c$  is the rigorous connectedness counterpart of that crossover: it is defined by the strict Euclidean pseudospectrum, and strict barrier decrease additionally excludes re-entrant splitting. Their convention  $A = -iH$  rotates the pseudospectrum by ninety degrees while leaving connectedness invariant.

## 7.2 Spectral-resolution classes at the threshold

The uncertainty considered here is deterministic, worst-case uncertainty in the Hamiltonian or model, distinct from a quantum uncertainty principle. Membership  $z \in \sigma_\varepsilon(A_n)$  records the existence of at least one perturbation of norm below  $\varepsilon$  that places an eigenvalue at  $z$ ; recovering the perturbation identity or a parent-branch label requires additional information, in the latter case a specified continuation path. Connected components nevertheless provide a precise, perturbation-stable coarse label. Define two unperturbed eigenvalues to be in the same  $\varepsilon$ -resolution class when they lie in the same component of  $\sigma_\varepsilon(A_n)$ . Along any continuous path  $\Delta A(s)$  satisfying  $\|\Delta A(s)\|_2 < \varepsilon$ , each continuous eigenvalue branch remains inside the pseudospectrum and, by continuity, within its initial component. Equivalently, each component carries a constant total algebraic multiplicity along such a path.

When a component contains one unperturbed eigenvalue, its resolution class gives an individual-mode label. Partial mergers may combine some labels earlier;  $N_c$  records the *last* merger. For  $n < N_c$ , at least two robust spectral classes remain, whereas

for  $n \geq N_c$  the sole component contains all  $n$  eigenvalues. Spectral location then supplies only the common component label. Knowledge of the actual perturbation or its continuation path may still permit eigenvalue-branch tracking.

Each fixed perturbation contributes its own  $n$  eigenvalues, while the connected pseudospectral region is assembled from points realized by different perturbations across the norm ball. The set therefore describes attainable spectral locations; a probability distribution requires an additional disorder ensemble. In resolvent language, for  $z \notin \sigma(A)$ ,

$$\|(zI_n - A)^{-1}\|_2 = \max_{\mathbf{f} \neq 0} \frac{\|\mathbf{u}\|_2}{\|\mathbf{f}\|_2}, \quad (zI_n - A)\mathbf{u} = \mathbf{f}.$$

Thus the single cloud identifies a region of large pointwise resolvent amplification at the specified worst-case resolution; each fixed noisy sample contributes only its own  $n$  spectral points to this union.

### 7.3 Scope and consequences

Finite-size merger also occurs beyond skin-effect settings. At the normal endpoint  $a = 1$ , increasingly dense Hermitian eigenvalues merge at fixed  $\varepsilon$  on the algebraic level-spacing scale. The exponential barrier and amplitude-skin-depth coefficient are distinctive features of the Hatano–Nelson family, and the corresponding fixed- $a$  asymptotic is nonuniform as  $a \uparrow 1$ .

The theorem concerns unstructured spectral-norm perturbations. Onsite, hopping, boundary, or Toeplitz restrictions can produce a different attainable set and merger threshold, requiring a structured-pseudospectral analysis. Pseudospectra of banded Toeplitz matrices are treated in [13], and Toeplitz-structured perturbations in [28]; robustness of a one-dimensional resonator skin effect under random imperfections in [29]; pseudospectral boundary sensitivity in correlated or disordered systems in [20]; and pseudospectral scaling and cloud merger in asymmetric lattices in [24]. Long-time dynamical amplification can also persist beyond the cloud-merger crossover [24].

Thus  $N_c$  is the exact threshold at which the clean open Hatano–Nelson chain loses robust separation into multiple pseudospectral resolution classes under unstructured spectral-norm uncertainty. This finite-size spectral-resolution transition is governed by the skin-depth scale; the onset of the skin effect, structured-disorder thresholds, individual noisy spectra, and dynamical amplification are governed by their corresponding observables and may occur at different scales.

## A Pentadiagonal continuant certificate for vertical monotonicity

**Proposition A.1** (Vertical determinant certificate). *Fix  $x \in \mathbb{R}$ ,  $Y, t \geq 0$ , and define  $\mathcal{P}_n(Y, t)$  and  $F_n(Y, t)$  as in the proof of lemma 3.1. For every  $n \geq 1$ ,*

$$\mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0.$$

*Proof.* At a fixed value of  $Y$ , write

$$d = x^2 + Y + 1 + a^2 - t^2, \quad b = -(1 + a)x + i(1 - a)\sqrt{Y}.$$

For  $n \geq 2$ , the matrix  $\mathcal{P}_n(Y, t)$  has diagonal

$$d - 1, d, \dots, d, d - a^2,$$

whereas for  $n = 1$  its sole entry is  $d - 1 - a^2$ . In all dimensions its first upper diagonal is  $b$ , its first lower diagonal is  $\bar{b}$ , and its second off-diagonals are  $a$ . Let  $T_k$  be the

determinant of the pure pentadiagonal Toeplitz matrix with diagonal  $d$  and these two off-diagonals. Set  $T_{-1} = 0$ ,  $T_0 = 1$ , and put

$$D_k = T_k - T_{k-1}, \quad R_k = T_k - a^2 T_{k-1}. \quad (\text{A.1})$$

Indeed, a determinant is affine in each diagonal entry. Replacing the first diagonal entry  $d$  by  $d - 1$  subtracts the cofactor  $T_{k-1}$ , whereas replacing the last one by  $d - a^2$  subtracts  $a^2 T_{k-1}$ . Thus, for  $k < n$ ,  $D_k$  and  $R_k$  are, respectively, the leading and trailing principal minors of  $\mathcal{P}_n(Y, t)$ .

The following continuant calculation isolates all dependence on  $Y$ . With  $T_k = 0$  for  $k < 0$ , a Laplace expansion over the last two rows, grouped according to the largest displaced index in the contributing permutation, gives for every  $k \geq 1$

$$\begin{aligned} T_k = & dT_{k-1} - (|b|^2 - a^2)T_{k-2} - (2a^2d - a(b^2 + \bar{b}^2))T_{k-3} \\ & - a^2(|b|^2 - a^2)T_{k-4} + a^4dT_{k-5} - a^6T_{k-6} - a^2\mathbf{1}_{\{k=2\}}. \end{aligned} \quad (\text{A.2})$$

For example,  $T_1 = d$  and  $T_2 = d^2 - |b|^2$ ; these also verify the only inhomogeneous coefficient in (A.2). Multiplying the recurrence by  $z^k$  and summing gives

$$\sum_{k \geq 0} T_k z^k = \frac{1 - a^2 z^2}{Q(z, Y)}, \quad (\text{A.3})$$

where

$$\begin{aligned} Q(z, Y) = & 1 - dz + (|b|^2 - a^2)z^2 + (2a^2d - a(b^2 + \bar{b}^2))z^3 \\ & + a^2(|b|^2 - a^2)z^4 - a^4dz^5 + a^6z^6. \end{aligned}$$

Consequently, with  $\mathcal{D}(z, Y) = \sum_{k \geq 0} D_k z^k$ ,

$$\mathcal{D}(z, Y) = \frac{(1 - z)(1 - a^2 z^2)}{Q(z, Y)}, \quad \sum_{k \geq 0} F_k(Y, t) z^k = (1 - a^2 z) \mathcal{D}(z, Y). \quad (\text{A.4})$$

For the second identity, first change the initial diagonal, producing  $D_k$ , and then change the final diagonal; its cofactor is  $D_{k-1}$ . Thus  $F_k = D_k - a^2 D_{k-1}$ , which proves the asserted generating function. Direct substitution of  $d$  and  $b$  gives

$$\partial_Y Q = -z(1 - z)(1 + az)^2(1 - a^2 z).$$

Differentiating (A.4) therefore yields

$$\sum_{k \geq 0} \partial_Y F_k(Y, t) z^k = \varphi_a(z) \mathcal{D}(z, Y)^2, \quad \varphi_a(z) = \frac{z(1 + az)(1 - a^2 z)^2}{1 - az}. \quad (\text{A.5})$$

Suppose that  $\mathcal{P}_n(Y, t) > 0$ . Then

$$D_k > 0, \quad R_k > 0 \quad (0 \leq k < n). \quad (\text{A.6})$$

Put  $u_k = \sum_{j=0}^k D_j D_{k-j}$ , with  $u_k = 0$  for  $k < 0$ . The coefficients of  $\varphi_a$ , beginning with the coefficient of  $z$ , are

$$1, \quad 2a(1 - a), \quad a^2(a^2 - 4a + 2), \quad 2a^{k-1}(1 - a)^2 \quad (k \geq 4).$$

Thus

$$\begin{aligned} \partial_Y F_n = & u_{n-1} + 2a(1 - a)u_{n-2} + a^2(a^2 - 4a + 2)u_{n-3} \\ & + 2(1 - a)^2 \sum_{k=4}^n a^{k-1} u_{n-k}. \end{aligned} \quad (\text{A.7})$$

The third coefficient is the sole term that may be negative; the following cofactor inequality controls its contribution. Let  $j, k \geq 1$ ,  $j + k = n - 1$ . Deleting index  $k + 1$  from  $\mathcal{P}_n$  leaves blocks of sizes  $k$  and  $j$ , with the second off-diagonal  $a$  as their sole coupling. In block form, the deleted principal submatrix is

$$\begin{bmatrix} L_k & a\mathbf{e}_k\mathbf{e}_1^T \\ a\mathbf{e}_1\mathbf{e}_k^T & R_j^{\text{mat}} \end{bmatrix}, \quad \det L_k = D_k, \quad \det R_j^{\text{mat}} = R_j.$$

Expanding once in each of the two coupling entries shows that its determinant is

$$D_k R_j - a^2 D_{k-1} R_{j-1} > 0. \quad (\text{A.8})$$

Deleting index  $j + 1$  instead gives the same inequality with  $j, k$  interchanged. Define

$$\vartheta_k = \frac{D_k}{D_{k-1}}, \quad \varpi_k = \frac{R_k}{R_{k-1}}.$$

Since  $T_k = \sum_{\ell=0}^k D_\ell$ , (A.1) gives

$$R_k = D_k + (1 - a^2) \sum_{\ell=0}^{k-1} D_\ell, \quad R_k - R_{k-1} = D_k - a^2 D_{k-1}. \quad (\text{A.9})$$

Hence  $\vartheta_k \leq a^2$  implies  $\varpi_k \leq 1$ . On dividing (A.8) and its companion by positive factors, we obtain

$$\vartheta_k \varpi_j > a^2, \quad \vartheta_j \varpi_k > a^2. \quad (\text{A.10})$$

If both  $\vartheta_j$  and  $\vartheta_k$  were at most  $a^2$ , then both  $\varpi_j$  and  $\varpi_k$  would be at most 1, contradicting (A.10). Hence, after interchanging  $j, k$  if necessary,  $\vartheta_j > a^2$ . Moreover,

$$\vartheta_j R_{j-1} - a^2 R_j = (\vartheta_j - a^2)(R_{j-1} - a^2 D_{j-1}),$$

and the second factor equals  $(1 - a^2) \sum_{\ell=0}^{j-1} D_\ell > 0$ . Therefore

$$\vartheta_j > a^2 \implies \varpi_j < \vartheta_j / a^2.$$

The first inequality in (A.10) now gives  $a^2 < \vartheta_k \varpi_j < \vartheta_j \vartheta_k / a^2$ , and therefore

$$\vartheta_j \vartheta_k > a^4. \quad (\text{A.11})$$

Consequently, for  $n \geq 3$ , applying (A.11) to  $j = p + 1$ ,  $k = n - 2 - p$  gives

$$D_{p+1} D_{n-2-p} > a^4 D_p D_{n-3-p}, \quad 0 \leq p \leq n - 3.$$

Summing and then adding the two positive endpoint terms gives

$$u_{n-1} = 2D_{n-1} + \sum_{j=1}^{n-2} D_j D_{n-1-j} > a^4 u_{n-3}. \quad (\text{A.12})$$

If the third coefficient in (A.7) is negative, write it as  $-\mathcal{B}_{\text{neg}}$ , where  $\mathcal{B}_{\text{neg}} = a^2(4a - 2 - a^2)$ . Since

$$a^4 - \mathcal{B}_{\text{neg}} = 2a^2(1 - a)^2 > 0,$$

equations (A.7) and (A.12) prove (3.1). If that coefficient is nonnegative, the implication is immediate. For  $n = 1$ , one has  $\partial_Y F_1 = 1$ ; for  $n = 2$ , one has  $\partial_Y F_2 = 2D_1 + 2a(1 - a) > 0$ . Thus (3.1) holds in every dimension, which proves the proposition.  $\square$

## B Folded-continuant transfer and boundary residuals

This appendix supplies the transfer and generating identities underlying (4.10)–(4.11) and records every boundary sign and parity convention in the signed-pencil argument. Pair the coordinates in the order  $(1, n), (2, n-1), \dots$ . In this order the symmetric matrix  $(xI_n - A_n)J_n - sI_n$  is block tridiagonal. Its interior diagonal and off-diagonal blocks are

$$D_{\text{int}} = \begin{bmatrix} -s & x \\ x & -s \end{bmatrix}, \quad E_{\text{off}} = \begin{bmatrix} 0 & -1 \\ -a & 0 \end{bmatrix},$$

with lower block  $E_{\text{off}}^T$ . For even order the terminal block is

$$D_{\text{term}}^e = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix};$$

for odd order the terminal block is the scalar  $x-s$ , coupled to the preceding block by the column  $(-1, -a)^T$ . Since

$$(xI_n - A_n - sJ_n) = ((xI_n - A_n)J_n - sI_n)J_n,$$

the determinant of this block matrix, multiplied by  $\det J_n$ , is  $q_n(x, s)$ . Set  $\mathcal{Q}_m^e = q_{2m}(x, s)$  and  $\mathcal{Q}_m^o = q_{2m+1}(x, s)$ . The following five-minor transfer completes the continuant verification. For  $\tau \in \{e, o\}$ , let  $K_m^\tau$  denote the corresponding matrix

$$(xI_n - A_n - sJ_n)J_n = (xI_n - A_n)J_n - sI_n, \quad n = 2m \text{ or } 2m+1,$$

after simultaneously permuting its rows and columns into the paired order. Thus  $K_m^e$  has order  $2m$ ,  $K_m^o$  has order  $2m+1$ , and

$$K_1^e = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix}, \quad K_1^o = \begin{bmatrix} -s & x & -1 \\ x & -s & -a \\ -1 & -a & x-s \end{bmatrix}.$$

For  $m \geq 1$ , increasing  $m$  by one adds one interior block at the exposed end:

$$K_{m+1}^\tau = \begin{bmatrix} D_{\text{int}} & \mathcal{E}_m^\tau \\ (\mathcal{E}_m^\tau)^T & K_m^\tau \end{bmatrix}, \quad \mathcal{E}_m^\tau = \begin{bmatrix} E_{\text{off}} & 0 \end{bmatrix}, \quad (\text{B.1})$$

where the zero block has the width required by  $K_m^\tau$ .

For a matrix whose first two coordinates are exposed, write  $K(\hat{I} | \hat{J})$  for the matrix obtained by deleting rows  $I$  and columns  $J$ . Suppressing the parity superscript, set

$$\begin{aligned} \mathfrak{d}_m &= \det K_m, & \mathfrak{p}_m &= \det K_m(\hat{1} | \hat{1}), & \mathfrak{q}_m &= \det K_m(\hat{2} | \hat{2}), \\ \mathfrak{c}_m &= \det K_m(\hat{1} | \hat{2}), & \mathfrak{r}_m &= \det K_m(\widehat{1, 2} | \widehat{1, 2}). \end{aligned}$$

The off-principal minor obtained by deleting row 2 and column 1 has the same value  $\mathfrak{c}_m$ , because  $K_m$  is symmetric.

The one-step transfer follows from a Schur complement. On the dense set where both  $K_m$  and the tail obtained by deleting its first two coordinates are invertible, let

$$F = \begin{bmatrix} u & w \\ w & v \end{bmatrix}$$

be its Schur complement in  $K_m$ , and let  $\delta$  be the determinant of that tail. Then

$$(\mathfrak{d}_m, \mathfrak{p}_m, \mathfrak{q}_m, \mathfrak{c}_m, \mathfrak{r}_m) = \delta(uv - w^2, v, u, w, 1).$$

Since

$$E_{\text{off}} \text{adj}(F) E_{\text{off}}^T = \begin{bmatrix} u & -aw \\ -aw & a^2v \end{bmatrix},$$

taking the Schur complement of  $K_m$  in (B.1) gives

$$\begin{aligned} \mathfrak{d}_{m+1} &= (s^2 - x^2)\mathfrak{d}_m + s(a^2\mathfrak{p}_m + \mathfrak{q}_m) - 2ax\mathfrak{c}_m + a^2\mathfrak{r}_m, \\ \mathfrak{p}_{m+1} &= -s\mathfrak{d}_m - a^2\mathfrak{p}_m, \\ \mathfrak{q}_{m+1} &= -s\mathfrak{d}_m - \mathfrak{q}_m, \\ \mathfrak{c}_{m+1} &= x\mathfrak{d}_m + a\mathfrak{c}_m, \\ \mathfrak{r}_{m+1} &= \mathfrak{d}_m. \end{aligned} \tag{B.2}$$

Both sides are polynomial in  $x$  and  $s$ . For each fixed  $0 < a < 1$ , the identities hold on a dense nonsingular set, so the polynomial identity theorem extends them to all  $x, s$ , including singular matrix cases.

The five quantities in (B.2) satisfy one linear invariant. Define

$$\ell_m = (1 - a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1 - a)x\mathfrak{c}_m - a(1 - a)\mathfrak{r}_m.$$

Direct substitution into (B.2) gives

$$\ell_{m+1} = -a\ell_m.$$

For the two terminal blocks, direct evaluation gives

$$\begin{aligned} &(\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_e \\ &= ((1 + s)(a + s) - x^2, -a - s, -1 - s, x, 1), \\ &(\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_o \\ &= (x(1 + a^2) + 2as - C_0(x - s), \\ &\quad s^2 - sx - a^2, s^2 - sx - 1, x^2 - sx - a, x - s). \end{aligned}$$

Thus  $\ell_1 = 0$  in both parity classes, and consequently

$$(1 - a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1 - a)x\mathfrak{c}_m - a(1 - a)\mathfrak{r}_m = 0 \quad (m \geq 1). \tag{B.3}$$

Because

$$\det J_{2m} = \det J_{2m+1} = (-1)^m,$$

the first coordinate of the following signed state is the required determinant:

$$\boldsymbol{\eta}_m^\tau = (-1)^m \begin{bmatrix} \mathfrak{d}_m^\tau & \mathfrak{p}_m^\tau & \mathfrak{q}_m^\tau & \mathfrak{c}_m^\tau \end{bmatrix}^T, \quad (\boldsymbol{\eta}_m^e)_1 = \mathcal{Q}_m^e, \quad (\boldsymbol{\eta}_m^o)_1 = \mathcal{Q}_m^o.$$

Eliminating  $\mathfrak{r}_m$  with (B.3) reduces (B.2) to the four-dimensional transfer

$$\boldsymbol{\eta}_{m+1}^\tau = \mathsf{T}_{\text{fold}} \boldsymbol{\eta}_m^\tau, \quad \mathsf{T}_{\text{fold}} = \begin{bmatrix} x^2 - s^2 - a & \frac{a^3s}{1-a} & -\frac{s}{1-a} & ax \\ s & a^2 & 0 & 0 \\ s & 0 & 1 & 0 \\ -x & 0 & 0 & -a \end{bmatrix}. \tag{B.4}$$

Set

$$\boldsymbol{\eta}_0^e = \begin{bmatrix} 1 & a^{-1} & 1 & 0 \end{bmatrix}^T, \quad \boldsymbol{\eta}_0^o = \begin{bmatrix} x - s & 1 & 1 & -1 \end{bmatrix}^T. \tag{B.5}$$

These values incorporate the terminal conditions at  $m = 0$ . Indeed, direct multiplication in (B.4) gives

$$\mathbf{T}_{\text{fold}} \boldsymbol{\eta}_0^e = \begin{bmatrix} x^2 - (1+s)(a+s) \\ a+s \\ 1+s \\ -x \end{bmatrix},$$

and

$$\mathbf{T}_{\text{fold}} \boldsymbol{\eta}_0^o = \begin{bmatrix} C_0(x-s) - x(1+a^2) - 2as \\ a^2 - s^2 + sx \\ 1 - s^2 + sx \\ a + sx - x^2 \end{bmatrix},$$

which are exactly the signed terminal minors above. Hence  $\boldsymbol{\eta}_m^\tau = \mathbf{T}_{\text{fold}}^m \boldsymbol{\eta}_0^\tau$  for every  $m \geq 0$ .

A direct determinant calculation gives

$$\begin{aligned} \det(\lambda I_4 - \mathbf{T}_{\text{fold}}) &= \lambda^4 - C_0 \lambda^3 + \mathfrak{b} \lambda^2 - a^2 C_0 \lambda + a^4, \\ \mathfrak{b} &= (1 + a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3. \end{aligned} \quad (\text{B.6})$$

Apply Cayley–Hamilton to  $\mathbf{T}_{\text{fold}}$ , multiply by  $\mathbf{T}_{\text{fold}}^{m-4} \boldsymbol{\eta}_0^\tau$ , and extract the first coordinate. For both  $Z_m = \mathcal{Q}_m^e$  and  $Z_m = \mathcal{Q}_m^o$ , this yields

$$Z_m - C_0 Z_{m-1} + \mathfrak{b} Z_{m-2} - a^2 C_0 Z_{m-3} + a^4 Z_{m-4} = 0, \quad m \geq 4. \quad (\text{B.7})$$

Finally, applying  $\mathbf{T}_{\text{fold}}$  three times to (B.5) gives the boundary residuals

$$\begin{aligned} \text{Res}_0 &= Z_0, \quad \text{Res}_1 = Z_1 - C_0 Z_0, \quad \text{Res}_2 = Z_2 - C_0 Z_1 + \mathfrak{b} Z_0, \\ \text{Res}_3 &= Z_3 - C_0 Z_2 + \mathfrak{b} Z_1 - a^2 C_0 Z_0, \end{aligned}$$

namely

|                         | Res <sub>0</sub> | Res <sub>1</sub>      | Res <sub>2</sub>    | Res <sub>3</sub> |
|-------------------------|------------------|-----------------------|---------------------|------------------|
| $Z_m = \mathcal{Q}_m^e$ | 1                | $a(1 - 2\zeta_s)$     | $a^2(1 - 2\zeta_s)$ | $a^3$            |
| $Z_m = \mathcal{Q}_m^o$ | $x - s$          | $-[x(1 + a^2) + 2as]$ | $a^2(x - s)$        | 0                |

Thus, as a formal-power-series identity,

$$\Pi(z) \sum_{m \geq 0} Z_m z^m = \text{Res}_0 + \text{Res}_1 z + \text{Res}_2 z^2 + \text{Res}_3 z^3.$$

For the even sequence the polynomial on the right factors as  $(1 + az)(1 - 2a\zeta_s z + a^2 z^2)$ ; for the odd sequence it is  $(x - s) - [x(1 + a^2) + 2as]z + a^2(x - s)z^2$ . These factorisations identify the boundary-residual polynomials. Since  $\Pi(0) = 1$ , division by  $\Pi$  in  $\mathbb{R}[[z]]$  gives the two literal formal-power-series identities

$$\sum_{m \geq 0} \mathcal{Q}_m^e z^m = (1 + az)(1 - 2a\zeta_s z + a^2 z^2) \Pi(z)^{-1}, \quad (\text{B.8})$$

and

$$\sum_{m \geq 0} \mathcal{Q}_m^o z^m = ((x - s) - [x(1 + a^2) + 2as]z + a^2(x - s)z^2) \Pi(z)^{-1}. \quad (\text{B.9})$$

### C Divided differences for the folded Chebyshev formulas

The transfer calculation in [appendix B](#) gives two rational generating functions. Coefficient extraction converts them to the divided-difference formulas used in the half-angle chord parametrization. Set  $U_{-1} = 0$ . The Chebyshev generating function gives

$$\sum_{m \geq 0} (u - \zeta_s)(U_m(u) + U_{m-1}(u))\omega_{\text{gf}}^m = \frac{(u - \zeta_s)(1 + \omega_{\text{gf}})}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2},$$

and, with  $\mathbf{B}_s^{\text{fold}} = [x(1 + a^2) + 2as]/(2a)$ ,

$$\sum_{m \geq 0} ((x - s)u - \mathbf{B}_s^{\text{fold}})U_m(u)\omega_{\text{gf}}^m = \frac{(x - s)u - \mathbf{B}_s^{\text{fold}}}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2}.$$

For a differentiable real function  $F$ , write

$$[F]_{u,v} := \begin{cases} \frac{F(u) - F(v)}{u - v}, & u \neq v, \\ F'(u), & u = v \end{cases}.$$

Let  $u, v \in \mathbb{R}$  satisfy the two symmetric relations in (4.9). Taking the corresponding divided difference of the Chebyshev generating identities and putting  $\omega_{\text{gf}} = az$  recovers (B.8) and (B.9). Coefficient comparison gives

$$q_{2m}(x, s) = a^m [\mathbf{f}_m]_{u,v}, \quad \mathbf{f}_m(t) = (t - \zeta_s)(U_m(t) + U_{m-1}(t)), \quad (\text{C.1})$$

and

$$q_{2m+1}(x, s) = a^m [\mathbf{g}_m]_{u,v}, \quad \mathbf{g}_m(t) = \left( (x - s)t - \frac{x(1 + a^2) + 2as}{2a} \right) U_m(t). \quad (\text{C.2})$$

The polynomial determinant recurrence and generating identities hold globally. Once the two symmetric relations have been verified for a real pair, the real-pair formulas (4.10)–(4.11) apply, with the displayed derivative providing the collision value. For  $\Delta_{\text{fold}} < 0$ , the polynomial identities apply directly, independently of any square-root branch.

### D Exhaustion of the endpoint-side chord sheet

**Lemma D.1** (Endpoint-side chord-sheet exhaustion). *Under the chord parametrization and branch conventions of [equations \(4.15\) and \(4.16\)](#), the two exhaustion statements made in the main text hold in every positive gap, including a gap adjacent to zero. For odd  $K$ , they also hold on the positive half of the single central gap.*

*Proof.* Continue the ordered middle root  $s_*(x)$  through the open gap. Write  $\Delta_*(x) = \Delta_{\text{fold}}(x, s_*(x))$  for its folded discriminant, let  $y(x) \leq z(x)$  be the reconstructed non-negative half variables on the real sheet issuing from the endpoint germ, and set  $\Theta(x) = \arccos y(x)$ . Let  $\mathcal{C}$  be the set of gap points for which

$$\Delta_*(x) > 0, \quad \frac{j\pi}{K} < \Theta(x) < \frac{(j+1)\pi}{K}, \quad u_K^{\text{max}} < z(x) < \mathbf{c}_h.$$

On  $\mathcal{C}$ , the reversible folded determinant identities give  $\mathcal{W}_K(y(x)) = \mathcal{W}_K(z(x))$ ; uniqueness on the endpoint-side outer lobe then gives

$$z(x) = z_K(\mathcal{W}_K(y(x))).$$

Thus  $\mathcal{C}$  is precisely the selected endpoint-side chord sheet. The endpoint germ constructed in the main text makes  $\mathcal{C}$  nonempty. The ordered eigenvalues of the continuous real symmetric pencil  $B_{K-1}(x)$  are continuous in  $x$ , including at an eigenvalue crossing, so the continuation passes through a multiple least singular value. The folded pair  $u = 2z^2 - 1$ ,  $v = 2y^2 - 1$  and its reconstructed half variables are continuous, while the sheet inequalities are strict. Thus  $\mathcal{C}$  is relatively open in the gap.

It is also relatively closed. At a relative closure point the chord identity and all sheet inequalities persist weakly. The possible closure cases are as follows. (1) If  $\Delta_*(x) = 0$ , then the reconstructed half variables coincide, which is impossible because

$$z \geq u_K^{\max} > \cos \frac{\pi}{K} \geq \cos \frac{j\pi}{K} \geq y.$$

(2) At an interior point of a positive gap, including one adjacent to zero, both half variables retain their signs:  $z > 0$ , while  $y = 0$  would force  $x = 0$ , outside the open positive half. (3) At an interior point the outer variable stays below  $\mathfrak{c}_h$ , because equality in (4.16) would give  $s = 0$ . (4) The inner variable stays away from a nodal endpoint: the common level would then be zero, the endpoint-side solution would equal  $\mathfrak{c}_h$ , and (4.16) would give  $s = 0$ , whereas the middle root is nonzero inside a spectral gap. (5) The outer variable remains in its endpoint-side component, since its first exit point would be  $u_K^{\max}$ , contradicting (4.19). Passage through  $z = 1$  is harmless; only its coordinate changes. Hence every relative closure point remains in  $\mathcal{C}$ . Since the open gap is an interval,  $\mathcal{C}$  is the entire gap. This proves the first exhaustion direction: every actual gap point lies on the selected endpoint-side chord sheet.

For the converse direction, the reconstructed angle  $\Theta$  extends continuously to the closed spectral gap. The exact folded variables at the two endpoints give

$$\Theta(\lambda_{K-1,j+1}) = \frac{(j+1)\pi}{K}, \quad \Theta(\lambda_{K-1,j}) = \frac{j\pi}{K}.$$

For every  $\theta_0 \in (j\pi/K, (j+1)\pi/K)$ , the intermediate value theorem therefore supplies an interior gap point  $x$  with  $\Theta(x) = \theta_0$ . The classification just proved and uniqueness of the endpoint-side outer solution then give

$$y(x) = \cos \theta_0, \quad z(x) = z_K(\mathcal{W}_K(\cos \theta_0)),$$

so the corresponding abscissa and signed root are exactly those of the prescribed  $j$ -chord. Thus every chord parameter produces an actual point of the ordered middle branch. By lemma 4.2, its positive magnitude is  $s_{\min}(xI_{K-1} - A_{K-1})$ . Both exhaustion directions follow from continuity and sheet classification; monotonicity of  $\theta \mapsto x$  is unnecessary.

When  $K$  is odd, the same argument applies to the positive half of the single central gap, continuing from the positive spectral endpoint to  $x = 0$ . Because  $z > 0$ , the identity  $x = (2r/\mathfrak{c}_h) \cos \theta z$  forces  $\theta = \pi/2$  at  $x = 0$ . Thus every parameter on that half occurs, and reflection covers the negative half.  $\square$

## E Scalar estimate for the lower central-gap comparison

This appendix proves the strict scalar inequality (4.35) through finite estimates valid uniformly over the three cases specified below.

Put  $b = u_+/u_- = 1/a$ , and define

$$Q_b = b^2 - 2b\rho + 1, \quad S_b = b^2 + 2b\omega_L + 1.$$

The Chebyshev identity gives  $u_-^2 Q_b = 1$ , and

$$\mathfrak{D}_L = S_b/Q_b, \quad \Delta_\xi := \rho - \xi = \Delta_\omega Q_b/S_b. \quad (\text{E.1})$$

Indeed,  $u_+ = bu_-$  and  $u_-^2 = Q_b^{-1}$ , so

$$\mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+ = 1 + \frac{2b(\rho + \omega_L)}{Q_b} = \frac{S_b}{Q_b},$$

and the formula for  $\Delta_\xi$  follows from (4.34). The ratio  $U_L/U_{L-1}$  is strictly increasing above the largest zero of  $U_{L-1}$ , and equals 1 at  $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$ . Hence  $\rho > \rho_{\text{ref}}$ . Since  $S_b - Q_b = 2b(\rho + \omega_L) > 0$ ,

$$\xi > \rho_{\text{ref}} - \Delta_\omega > 2\omega_L - 1 > 2\omega_L^2 - 1 = \cos \frac{\pi}{L}.$$

Therefore  $f_L(x) = (1+x)U_{L-1}(x)$  is positive, strictly increasing, and strictly log-concave on  $[\xi, \rho]$ . In particular,  $\mathfrak{d}_\rho = (\log f_L)'(\rho) > 0$ , and

$$\mathfrak{d}_\rho = \begin{cases} \frac{L(b-\rho)-1}{\rho^2-1}, & \rho \neq 1, \\ \frac{2L^2+1}{6}, & \rho = 1 \end{cases}.$$

The endpoint value follows directly from  $U_{L-1}(1) = L$  and  $U'_{L-1}(1) = L(L^2 - 1)/3$ . Moreover,

$$\log f_L(\rho) - \log f_L(\xi) > \Delta_\xi \mathfrak{d}_\rho, \quad f_L(\xi)^2 < f_L(\rho)^2 e^{-2\Delta_\xi \mathfrak{d}_\rho} < \frac{f_L(\rho)^2}{1 + 2\Delta_\xi \mathfrak{d}_\rho}.$$

Here the first strict inequality is the integral form of strict log-concavity. Since  $f_L(\rho)^2 = (1+\rho)^2/Q_b$  and  $\Delta_\xi = \Delta_\omega Q_b/S_b$ , substitution into (4.35) reduces it to

$$\mathcal{N} = (\rho + \omega_L)(S_b + 2\Delta_\omega Q_b \mathfrak{d}_\rho) - (1 + \omega_L)(1 + \rho)^2 > 0. \quad (\text{E.2})$$

All three cases use

$$\Delta_\omega = 1 - \cos \frac{\pi}{2L} < \frac{\pi^2}{8L^2} < \frac{5}{4L^2}, \quad \Delta_\omega L < 1. \quad (\text{E.3})$$

Suppose first that  $\rho > 1$ . Put  $u = b - \rho$  and  $\Delta_+ = \rho^2 - 1$ . Since  $\mathfrak{d}_\rho > 0$ ,  $u > 1/L$ . Keeping  $\rho$  fixed, direct differentiation gives

$$\frac{\partial_b \mathcal{N}}{2(\rho + \omega_L)} = \rho + \omega_L + u - \Delta_\omega L + \frac{\Delta_\omega(3Lu^2 - 2u)}{\Delta_+}.$$

The right-hand side is strictly increasing for  $u \geq 1/L$  and is positive at  $u = 1/L$  because  $\Delta_\omega L < 1$ . Hence  $\mathcal{N}$  exceeds its value at  $b = \rho + 1/L$ . With  $s_0 = \rho + \omega_L$  and  $\omega_+ = 1 + \omega_L$ , that value is

$$\Upsilon(s_0) = s_0 \left[ \left( s_0 + \frac{1}{L} \right)^2 + \omega_+ \Delta_\omega \right] - \omega_+(s_0 + \Delta_\omega)^2.$$

For  $s_0 \geq \omega_+$ ,

$$\Upsilon''(s_0) = 6s_0 + 4/L - 2\omega_+ > 0, \quad \Upsilon'(\omega_+) = 2\omega_L \omega_+ + 4\omega_+/L + 1/L^2 > 0,$$

and

$$\Upsilon(\omega_+) = \omega_+(-2\Delta_\omega + 2\omega_+/L + 1/L^2) > 0.$$

Thus (E.2) holds when  $\rho > 1$ . If  $\rho = 1$ , then  $b = (L + 1)/L$ ,  $Q_b = L^{-2}$ , and direct substitution, using the endpoint value of  $\mathfrak{d}_\rho$ , gives

$$\mathcal{N} = \Upsilon(1 + \omega_L) + \frac{2(1 + \omega_L)\Delta_\omega}{L^2} \frac{2L^2 + 1}{6} > 0.$$

The first term is positive by the preceding calculation, and every factor in the correction term is positive.

For  $\rho < 1$ , set

$$v = 1 - \rho, \quad \Delta_b = b - 1, \quad u = \Delta_b + v, \quad \Delta_- = v(2 - v).$$

Writing  $\rho = \cos \theta_\rho$  gives  $0 < L\theta_\rho < \pi/2$  and

$$u = b - \rho = \frac{\sin((L + 1)\theta_\rho)}{\sin(L\theta_\rho)} - \cos \theta_\rho = \sin \theta_\rho \cot(L\theta_\rho).$$

Because  $b > 1$ ,  $u = \Delta_b + v > v > 0$ . Also  $\tan(L\theta_\rho) > L \tan \theta_\rho > L \sin \theta_\rho$ , since  $x \mapsto \tan x/x$  is strictly increasing on  $(0, \pi/2)$ . Hence

$$0 < v < u < 1/L. \quad (\text{E.4})$$

With  $\rho$  fixed, write  $\mathcal{N}_\rho^<(\cdot)$  for  $\mathcal{N}$  viewed as a function of  $\Delta_b = b - 1$ . A direct expansion of (E.2) gives

$$\mathcal{N}_\rho^<(\Delta_b) - \mathcal{N}_\rho^<(0) = \frac{\Delta_b(\rho + \omega_L)}{\Delta_-} \mathfrak{q}(u),$$

where

$$\mathfrak{q}(u) = \Delta_- [2(2 - \Delta_\omega) + u - v] + 2\Delta_\omega [u + v - L(u^2 + uv + v^2) - L\Delta_-].$$

The function  $\mathfrak{q}$  is a concave quadratic. Moreover,

$$\mathcal{N}_\rho^<(0) = \frac{v}{2 - v} \left[ (2 - \Delta_\omega)(2 - v)^2 - 2\Delta_\omega^2 - 4\Delta_\omega L(2 - v - \Delta_\omega) \right].$$

The following elementary bounds hold; the second follows from  $\rho > \rho_{\text{ref}}$  and  $1 - \rho_{\text{ref}} < \pi^2/[2(2L + 1)^2] < 5/(2L + 1)^2$ :

$$\Delta_\omega < \frac{5}{4L^2}, \quad v < \frac{5}{(2L + 1)^2} \leq \frac{1}{5}. \quad (\text{E.5})$$

Although  $5/(2L + 1)^2 = 1/5$  when  $L = 2$ , the preceding strict bound still gives  $v < 1/5$ . These bounds show that the bracketed expression is positive, since it equals

$$(2 - v) \left[ (2 - \Delta_\omega)(2 - v) - 4\Delta_\omega L \right] + 2\Delta_\omega^2(2L - 1)$$

and, using  $L \geq 2$ ,

$$(2 - \Delta_\omega)(2 - v) > \frac{27}{16} \frac{9}{5} = \frac{243}{80} > \frac{5}{2} \geq 4\Delta_\omega L.$$

At the endpoints of (E.4),

$$\mathfrak{q}(v) = v \left[ (2 - v)(4 - 2\Delta_\omega) + 4\Delta_\omega - 4\Delta_\omega L(1 + v) \right] > 0,$$

$$\mathbf{q}(1/L) = v \left[ (2 - v)(4 - 2\Delta_\omega + 1/L - v) - 4\Delta_\omega L \right] > 0.$$

At the two endpoints, the positive leading terms exceed

$$\frac{9}{5} \left( 4 - \frac{5}{8} \right) = \frac{243}{40}, \quad \frac{9}{5} \left( 4 - \frac{5}{8} - \frac{1}{5} \right) = \frac{1143}{200},$$

while the negative terms have magnitudes less than  $(5/L)(6/5) \leq 3$  and  $5/L \leq 5/2$ , respectively. Concavity therefore gives  $\mathbf{q}(u) > 0$ , proving (E.2), and hence (4.35), in all three cases.

## F Signed-root certification in the lower central gap

This appendix verifies the folded substitution and second-singular-value separation used in the main proof of  $d_m > c_{m+1}$ .

At  $x = x_*$  and  $s = \pm c$ , the real pair  $u = \xi$ ,  $v = -\omega_L$  satisfies the two symmetric relations in (4.9). Indeed, its sum and product are  $\xi - \omega_L$  and  $-\omega_L \xi$ , and the required identities follow from

$$\mathfrak{D}_L = \frac{\zeta + \omega_L}{\zeta - \rho}, \quad \xi = \frac{\rho(\zeta + 1) + \zeta(\omega_L - 1)}{\zeta + \omega_L}.$$

Positivity of  $\xi + \omega_L$  selects the distinct-variable branch of the divided difference in (4.11). For convenience, write the folded numerator as

$$\mathbf{n}_s(t) = -[x(\zeta - t) + s(1 + t)]U_{L-1}(t).$$

Put

$$\mathbf{X}_\xi = x_*(\zeta - \xi), \quad \mathbf{Y}_\xi = c(1 + \xi), \quad \mathbf{X}_\omega = x_*(\zeta + \omega_L), \quad \mathbf{Y}_\omega = c\Delta_\omega,$$

and set  $U_\xi = U_{L-1}(\xi)$ ,  $U_{-\omega} = U_{L-1}(-\omega_L)$ . The identities

$$\mathfrak{D}_L = 2a(\zeta + \omega_L)u_+^2, \quad \mathfrak{D}_L(\rho + \omega_L)(\zeta - \xi) = (\zeta + \omega_L)(1 + \xi)$$

give  $\mathbf{Y}_\omega \mathbf{Y}_\xi = \mathbf{X}_\omega \mathbf{X}_\xi$ . The identity  $u_-^2 Q_b = 1$  gives  $Q_b > 0$ , and  $S_b - Q_b = 2b(\rho + \omega_L) > 0$ , so  $\mathfrak{D}_L = S_b/Q_b > 1$ . Moreover,  $L \geq 2$  gives  $\omega_L > 1/2$ , while  $\rho > \rho_{\text{ref}} > \omega_L$ ; hence

$$\rho + \omega_L > 2\omega_L > 1 > 1 - \omega_L > 0.$$

Consequently,

$$\frac{\mathbf{X}_\omega^2}{\mathbf{Y}_\omega^2} = \frac{\mathfrak{D}_L(\rho + \omega_L)}{1 - \omega_L} > 1,$$

so, with  $\mathbf{r}_* = \mathbf{X}_\xi/\mathbf{Y}_\xi = \mathbf{Y}_\omega/\mathbf{X}_\omega \in (0, 1)$ , the common factor  $a^{L-1}/(\xi + \omega_L)$  in (4.11) is positive, and the two numerator differences are

$$\begin{aligned} \mathbf{n}_c(\xi) - \mathbf{n}_c(-\omega_L) &= (1 + \mathbf{r}_*)(\mathbf{X}_\omega U_{-\omega} - \mathbf{Y}_\xi U_\xi), \\ \mathbf{n}_{-c}(\xi) - \mathbf{n}_{-c}(-\omega_L) &= (1 - \mathbf{r}_*)(\mathbf{X}_\omega U_{-\omega} + \mathbf{Y}_\xi U_\xi). \end{aligned}$$

Their product is positive by (4.35), because

$$U_{-\omega}^2 = \frac{1}{1 - \omega_L^2}, \quad \mathbf{X}_\omega^2 U_{-\omega}^2 = \frac{c^2 \mathfrak{D}_L(\rho + \omega_L)}{1 + \omega_L} > \mathbf{Y}_\xi^2 U_\xi^2.$$

Thus

$$q_{2L-1}(x_*, c)q_{2L-1}(x_*, -c) > 0. \tag{F.1}$$

At  $x = 0$ , the second-smallest singular value of  $A_{2L-1}$  is

$$s_{L,2}^{(0)} = \sqrt{1 + a^2 - 2a \cos \frac{\pi}{L}}.$$

Indeed, after separating odd and even coordinates, the two rectangular off-diagonal blocks have Gram matrices equal to the  $(L-1)$ -dimensional tridiagonal Toeplitz matrix with diagonal  $1 + a^2$  and off-diagonal  $a$ . Their singular values therefore have squares

$$1 + a^2 + 2a \cos \frac{k\pi}{L}, \quad 1 \leq k \leq L-1,$$

and each occurs twice, while the remaining singular value is zero. We claim

$$x_* + c < 2\sqrt{a} \sin \frac{\pi}{2L} \leq s_{L,2}^{(0)}. \quad (\text{F.2})$$

Indeed, recall  $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$ . Since  $\rho > \rho_{\text{ref}}$  and  $U_{L-1}$  is strictly increasing above its largest zero,

$$u_- = U_{L-1}(\rho) > U_{L-1}(\rho_{\text{ref}}) =: u_{-, \text{ref}}, \quad u_{-, \text{ref}}^{-2} = 2(1 - \rho_{\text{ref}}).$$

The identities  $c = a/u_-$  and  $c^2 = 2a(\zeta - \rho)$  imply  $\zeta - \rho < a(1 - \rho_{\text{ref}}) < 1 - \rho_{\text{ref}}$ . Also

$$\frac{x_*^2}{2a} = \Delta_\omega \frac{\rho + \omega_L}{\zeta + \omega_L} < \Delta_\omega.$$

Put

$$\theta_L = \frac{\pi}{4L}, \quad \Delta_L = \frac{\theta_L}{2L+1}.$$

Then  $\Delta_\omega = 2 \sin^2 \theta_L$  and  $1 - \rho_{\text{ref}} = 2 \sin^2(\theta_L - \Delta_L)$ . Since all quantities are positive, the preceding quotient bound for  $x_*^2$  gives

$$x_* < 2\sqrt{a} \sin \theta_L.$$

Likewise,

$$c^2 = 2a(\zeta - \rho) < 2a^2(1 - \rho_{\text{ref}}) = 4a^2 \sin^2(\theta_L - \Delta_L),$$

and hence

$$c < 2a \sin(\theta_L - \Delta_L) < 2\sqrt{a} \sin(\theta_L - \Delta_L),$$

where the last inequality uses  $0 < a < 1$ , hence  $a < \sqrt{a}$ . It suffices to prove

$$\sin \theta_L + \sin(\theta_L - \Delta_L) < \sin(2\theta_L). \quad (\text{F.3})$$

The estimates

$$2 \sin \theta_L - \sin(2\theta_L) = 2 \sin \theta_L (1 - \cos \theta_L) < \theta_L^3$$

and

$$\sin \theta_L - \sin(\theta_L - \Delta_L) = \int_{\theta_L - \Delta_L}^{\theta_L} \cos u \, du > \Delta_L \cos \theta_L$$

reduce (F.3) to  $\Delta_L \cos \theta_L > \theta_L^3$ . Substituting  $\Delta_L = \theta_L/(2L+1)$  and  $\theta_L = \pi/(4L)$  gives the equivalent condition  $2 \cos \theta_L > \theta_L(\pi + 2\theta_L)$ . For  $0 < \theta_L \leq \pi/8$ , using  $\pi^2 < 10$ ,

$$2 \cos \theta_L > 2 - \theta_L^2 \geq \frac{59}{32} > \frac{25}{16} \geq \theta_L(\pi + 2\theta_L).$$

Consequently  $x_* + c < 2\sqrt{a} \sin \frac{\pi}{2L}$ . Finally,

$$(s_{L,2}^{(0)})^2 - 4a \sin^2 \frac{\pi}{2L} = (1 - a)^2 \geq 0,$$

which proves (4.38).

## Declarations

Removed for anonymity.

## References

- [1] N. Hatano and D. R. Nelson. Localization transitions in non-Hermitian quantum mechanics. *Phys. Rev. Lett.*, 77:570–573, 1996.
- [2] N. Hatano and D. R. Nelson. Vortex pinning and non-Hermitian quantum mechanics. *Phys. Rev. B*, 56:8651–8673, 1997.
- [3] L. N. Trefethen and M. Embree. *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press, Princeton, NJ, 2005.
- [4] L. N. Trefethen. Pseudospectra of linear operators. *SIAM Rev.*, 39:383–406, 1997.
- [5] N. Okuma, K. Kawabata, K. Shiozaki, and M. Sato. Topological origin of non-Hermitian skin effects. *Phys. Rev. Lett.*, 124:086801, 2020.
- [6] S. Yao and Z. Wang. Edge states and topological invariants of non-Hermitian systems. *Phys. Rev. Lett.*, 121:086803, 2018.
- [7] K. Yokomizo and S. Murakami. Non-Bloch band theory of non-Hermitian systems. *Phys. Rev. Lett.*, 123:066404, 2019.
- [8] X. Zhang, T. Zhang, M.-H. Lu, and Y.-F. Chen. A review on non-Hermitian skin effect. *Adv. Phys.: X*, 7:2109431, 2022.
- [9] P. Schmidt and F. Spitzer. The Toeplitz matrices of an arbitrary Laurent polynomial. *Math. Scand.*, 8:15–38, 1960.
- [10] A. Böttcher, S. M. Grudsky, and B. Silbermann. Norms of inverses, spectra, and pseudospectra of large truncated Wiener–Hopf operators and Toeplitz matrices. *New York J. Math.*, 3:1–31, 1997.
- [11] L. Reichel and L. N. Trefethen. Eigenvalues and pseudo-eigenvalues of Toeplitz matrices. *Linear Algebra Appl.*, 162–164:153–185, 1992.
- [12] A. Böttcher, M. Embree, and L. N. Trefethen. Piecewise continuous Toeplitz matrices and operators: Slow approach to infinity. *SIAM J. Matrix Anal. Appl.*, 24:484–489, 2002.
- [13] A. Böttcher and S. M. Grudsky. *Spectral Properties of Banded Toeplitz Matrices*. SIAM, Philadelphia, 2005.
- [14] S. Noschese, L. Pasquini, and L. Reichel. Tridiagonal Toeplitz matrices: Properties and novel applications. *Numer. Linear Algebra Appl.*, 20:302–326, 2013.
- [15] R. Alam, S. Bora, R. Byers, and M. L. Overton. Characterization and construction of the nearest defective matrix via coalescence of pseudospectral components. *Linear Algebra Appl.*, 435:494–513, 2011.
- [16] J. V. Burke, A. S. Lewis, and M. L. Overton. Optimization and pseudospectra, with applications to robust stability. *SIAM J. Matrix Anal. Appl.*, 25:80–104, 2003.

- [17] A. S. Lewis and C. H. J. Pang. Variational analysis of pseudospectra. *SIAM J. Optim.*, 19:1048–1072, 2008.
- [18] L. N. Trefethen, M. Contedini, and M. Embree. Spectra, pseudospectra, and localization for random bidiagonal matrices. *Comm. Pure Appl. Math.*, 54:595–623, 2001.
- [19] N. Okuma and M. Sato. Hermitian zero modes protected by nonnormality: Application of pseudospectra. *Phys. Rev. B*, 102:014203, 2020.
- [20] N. Okuma and M. Sato. Non-Hermitian skin effects in Hermitian correlated or disordered systems: Quantities sensitive or insensitive to boundary effects and pseudo-quantum-number. *Phys. Rev. Lett.*, 126:176601, 2021.
- [21] H. Ammari, S. Barandun, J. Cao, B. Davies, and E. O. Hiltunen. Mathematical foundations of the non-Hermitian skin effect. *Arch. Ration. Mech. Anal.*, 248:33, 2024.
- [22] H. Ammari, S. Barandun, Y. De Bruijn, P. Liu, and C. Thalhammer. Spectra and pseudo-spectra of tridiagonal  $k$ -Toeplitz matrices and the topological origin of the non-Hermitian skin effect. *J. Phys. A: Math. Theor.*, 58:205201, 2025.
- [23] H. Ammari, S. Barandun, Y. De Bruijn, P. Liu, and C. Thalhammer. Corrigendum: Spectra and pseudo-spectra of tridiagonal  $k$ -Toeplitz matrices and the topological origin of the non-Hermitian skin effect (2025 J. Phys. A: Math. Theor. 58 205201). *J. Phys. A: Math. Theor.*, 59:079501, 2026.
- [24] I. Kiorpelidis and K. G. Makris. Scaling of pseudospectra in exponentially sensitive lattices. *Phys. Rev. Research*, 7:L032043, 2025.
- [25] J. Sirker. Pseudospectral phenomena and the origin of the non-Hermitian skin effect. preprint, arXiv:2603.22643v3 [cond-mat.stat-mech], 2026.
- [26] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth. On the Lambert  $W$  function. *Adv. Comput. Math.*, 5:329–359, 1996.
- [27] N. Hatano and D. R. Nelson. Non-Hermitian delocalization and eigenfunctions. *Phys. Rev. B*, 58:8384–8390, 1998.
- [28] P. Buttà, N. Guglielmi, and S. Noschese. Computing the structured pseudospectrum of a Toeplitz matrix and its extreme points. *SIAM J. Matrix Anal. Appl.*, 33:1300–1319, 2012.
- [29] H. Ammari, S. Barandun, B. Davies, E. O. Hiltunen, and P. Liu. Stability of the non-Hermitian skin effect in one dimension. *SIAM J. Appl. Math.*, 84:1697–1717, 2024.